



SpectreRF Workshop

Power Amplifier Design Using SpectreRF

MMSIM6.0USR2

November 2005

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Power Amplifier Design Measurements

The procedures described in this workshop are deliberately broad and generic. Your specific design might require procedures that are slightly different from those described here.

Purpose

This workshop describes how to use SpectreRF in the Analog Design Environment to measure parameters which are important in design verification of Power Amplifiers, or PAs. New features of MMSIM6.0USR2 are included.

Audience

Users of SpectreRF in the Analog Design Environment.

Overview

This application note describes a basic set of the most useful measurements for PAs.

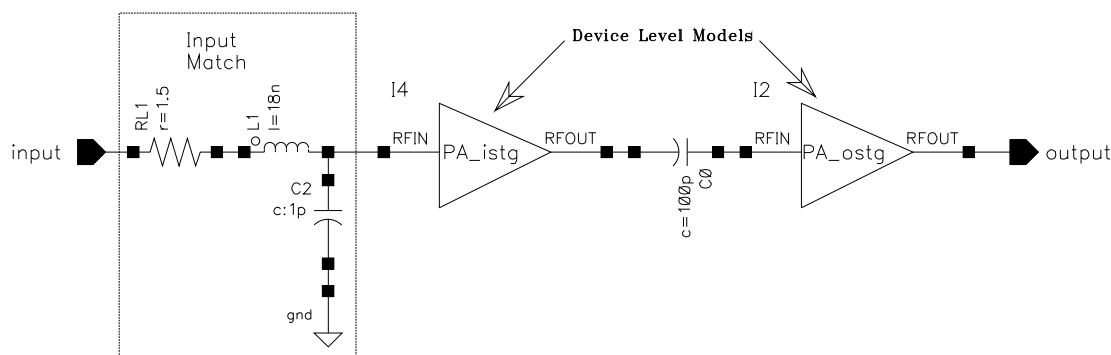
Introduction to Power Amplifiers

Power amplifiers, or PAs, are a part of the transmitter front-end used to amplify the transmitted signal so the signal can be received and decoded within a fixed geographical area. The main PA performance parameter is the output power level the PA can achieve, depending on the targeted application, linearity, and efficiency.

Power amplifiers can be categorized several ways depending on whether they are broadband or narrowband, and whether they are intended for linear operation (Class A, B, AB and C) or constant-envelope operation (Class D, E and F). This application note focuses on the design of narrowband and linear PAs.

The Design Example

The PA measurements described in this workshop are calculated using SpectreRF in the Analog Design Environment. The design example used to conduct the measurements described in this workshop is a two-stage power amplifier namely, *EF_PA_istg* and *EF_PA_ostg* as shown below:



The supply voltage is 5 V. There is a simple output matching network in the subcircuit *EF_PA_ostg*. The power amplifier is designed to be driven by CDMA I/Q channel baseband signals, modulated using QPSK schemes with a carrier frequency of 1 GHz.

Typical PA performance metrics as listed in following table:

Measurement	Acceptable Value
Output Power	+20 to +30 dBm
Efficiency	30% to 60%
Supply Voltage	2.8 to 5.8 V
Gain	20 to 30 dB
Harmonic Output (2f, 3f ,4f)	-30 to -50 dBc
Stability Factor	>1

Three testbenches for PA measurements

Testbench One

The first testbench is the PA driven by sinusoidal sources. Use this testbench to make the general measurements, which include

- Power related measurements (input power, output power, supply voltage, supply current, power gain and power dissipation)
- Efficiency measurements (drain efficiency and power added efficiency)
- Linearity measurements (1 dB compression point, IIP3 and OIP3)
- Noise measurements (NF or F)
- Stability measurements (K-factors, B1f and S-parameter)
- Large signal S-Parameter measurements.

Use the Periodic Steady State (PSS) analysis followed by the Periodic Small Signal (PAC/PSP/PNOISE) analyses to make these measurements. (See Lab1 to Lab4 from page 6 to page 45 for details.)

Testbench Two

The second testbench is the PA driven by a sinusoidal source with a port adapter added at the output power amplifier. Use this testbench to generate

- Load-pull contours
- Reflection contours

Use the swept PSS analysis combined with the parametric analysis tools to measure load pull. (See Lab5 on page 46 for details)

Testbench Three

The third testbench is the PA driven by modulation signals. It is used to generate

- ACPR plots
- Input and output trajectory plots

Use the Envelope Following (envlp) analysis to make these measurements. (See Lab6 on page 55 for details).

Example Measurements Using SpectreRF

To achieve optimal circuit performance, you should measure and evaluate several PA characteristics or parameters under varying conditions. The most important trade-off in PA design is between efficiency and linearity.

We'll begin our examination of the flow by bringing up the Cadence Design Framework II environment and look at a full view of our reference design:

Change directory to...

Action: cd to **./paSimple** directory

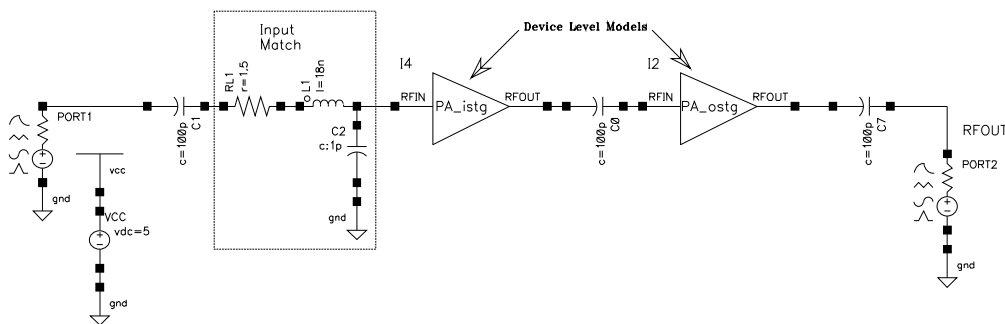
Action: Invoke tool **icfb&**

Action: In the CIW window, select **Tools->Library Manager...**

Lab1: Power Related Measurement (Swept PSS)

Power related measurements include input power, output power, supply voltage, supply current, power gain and power dissipation. To make these measurements, use a swept PSS analysis to sweep the input power level.

Action1-1: Open the *schematic* view of the design *EF_example_simple* in the library *RFworkshop*



Action1-2: Select the **PORT1** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

Parameter	Value
Resistance	50 ohm
Port Number	1
DC voltage	(blank)
Source type	sine
Frequency name 1	RF
Frequency 1	fin
Amplitude 1 (dBm)	pin

Action1-3: Select the **PORT2** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

Parameter	Value
Resistance	50 ohm
Port Number	2
DC voltage	(blank)
Source type	dc

Action1-4: Check and save the schematic.

Action1-5: In the Virtuoso Schematic Editing window, select **Tools->Analog Environment**

Action1-6: You can choose **Session—Load State** in Virtuoso Analog Design Environment load state “**Lab1_Power_PSS**”, then skip to [Action1-12](#) or ...

Action1-7: In Analog Design Environment window, select **Analyses->Choose...**

Action1-8: In the Choosing Analyses window, select the **pss** button in the **Analysis** field of the window. Configure the form as follows:

PA Design Using SpectreRF

Choosing Analyses – Virtuoso® Analog Design Environment

OK Cancel Defaults Apply Help

Engine ☐ Shooting ☐ Flexible Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
2	RF	fin	16	Large	PORT1

Large

Clear/Add Delete Update From Schematic

◆ Beat Frequency 16 Auto Calculate ■

◇ Beat Period

Output harmonics

Number of harmonics 40

Accuracy Defaults (emmpreset)

☐ conservative ■ moderate ☐ liberal

Additional Time for Stabilization (tstab)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Oscillator ☐

Sweep ■ Frequency Variable? ◆ no ◇ yes

Variable

Variable Name pin

Select Design Variable

Sweep Range

◆ Start-Stop Start -30 Stop 10

◇ Center-Span

Sweep Type

◆ Linear ◇ Logarithmic

◆ Step Size 5

◇ Number of Steps

Add Specific Points ☐

Enabled ■ Options...

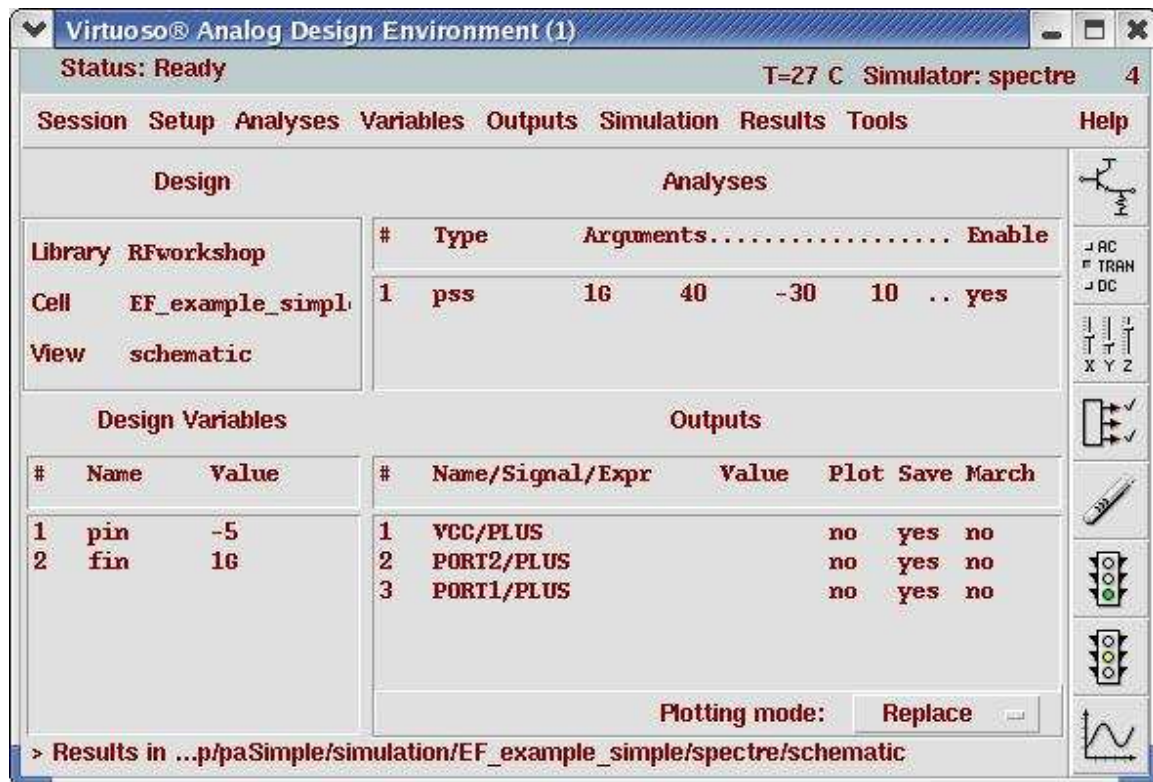
PA Design Using SpectreRF

Action1-9: Make sure that the Enabled button is on. Click OK in the Choosing Analyses form.

Action1-10: In the Virtuosos Analog Design Environment window, **Choose Outputs—To be Saved—Select on Schematic**

Action1-11: In the schematic, select the positive terminals of PORT2, PORT1 and VCC. Press **ESC** key to escape the selection process.

Now your Virtuoso Analog Design Environment looks like this:



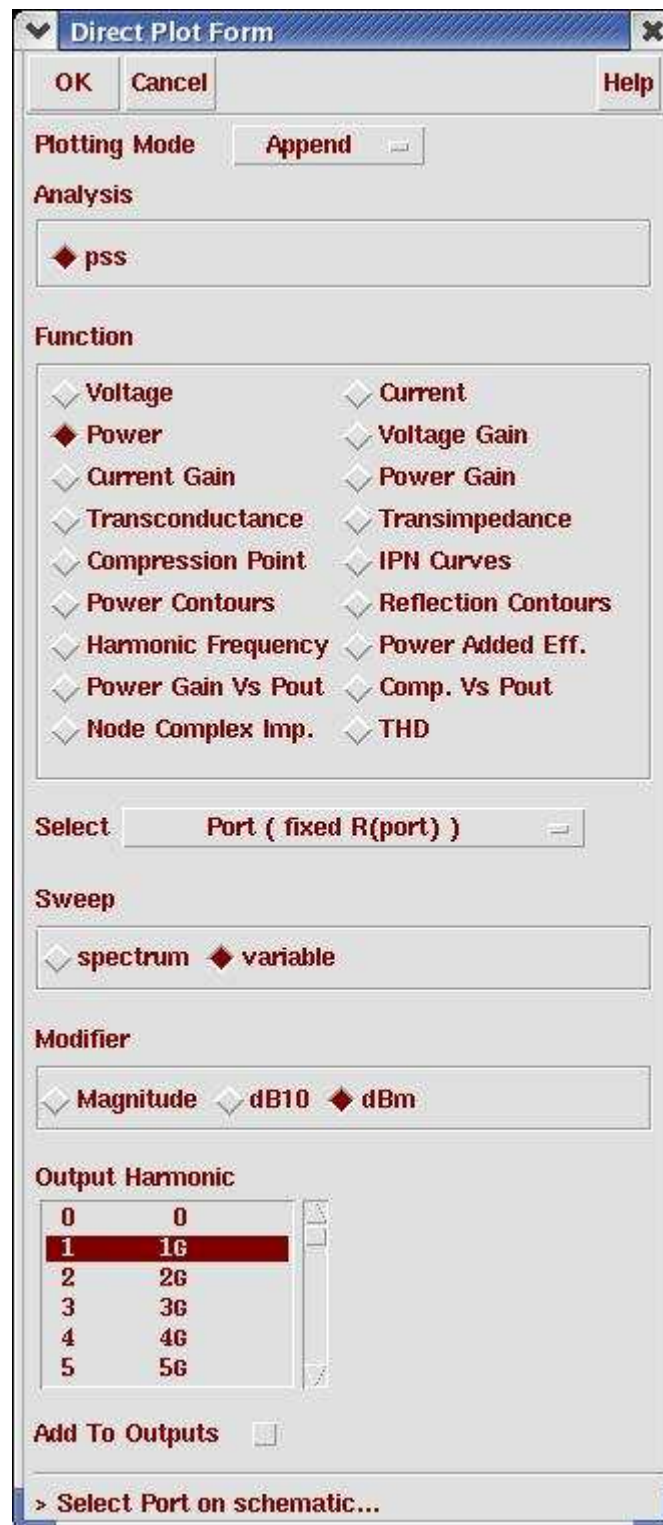
Action1-12: Choose **Simulation—Netlist and Run** to start the simulation or click on the **netlist and Run** icon in the Virtuoso Analog Design Environment window.

After the simulation has finished, plot the simulation results.

Action1-13: In the Virtuoso Analog Design Environment window, choose **Results—Direct Plot—Main Form**

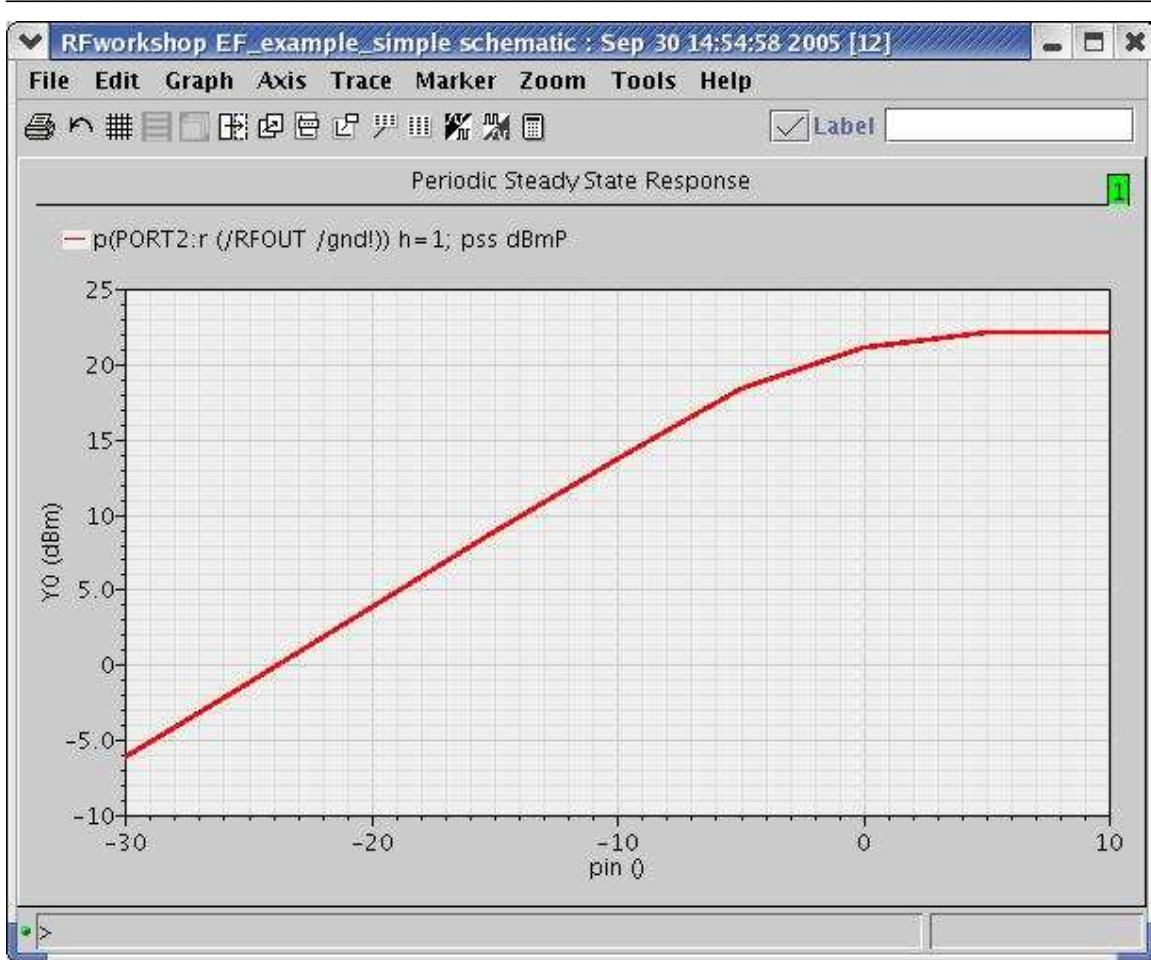
The Direct Plot form appears.

Action1-14: In Direct Plot Form window, choose **pss** as the Analysis type. Choose **Power** in Function field. Choose **1G** in the Output harmonic list box.



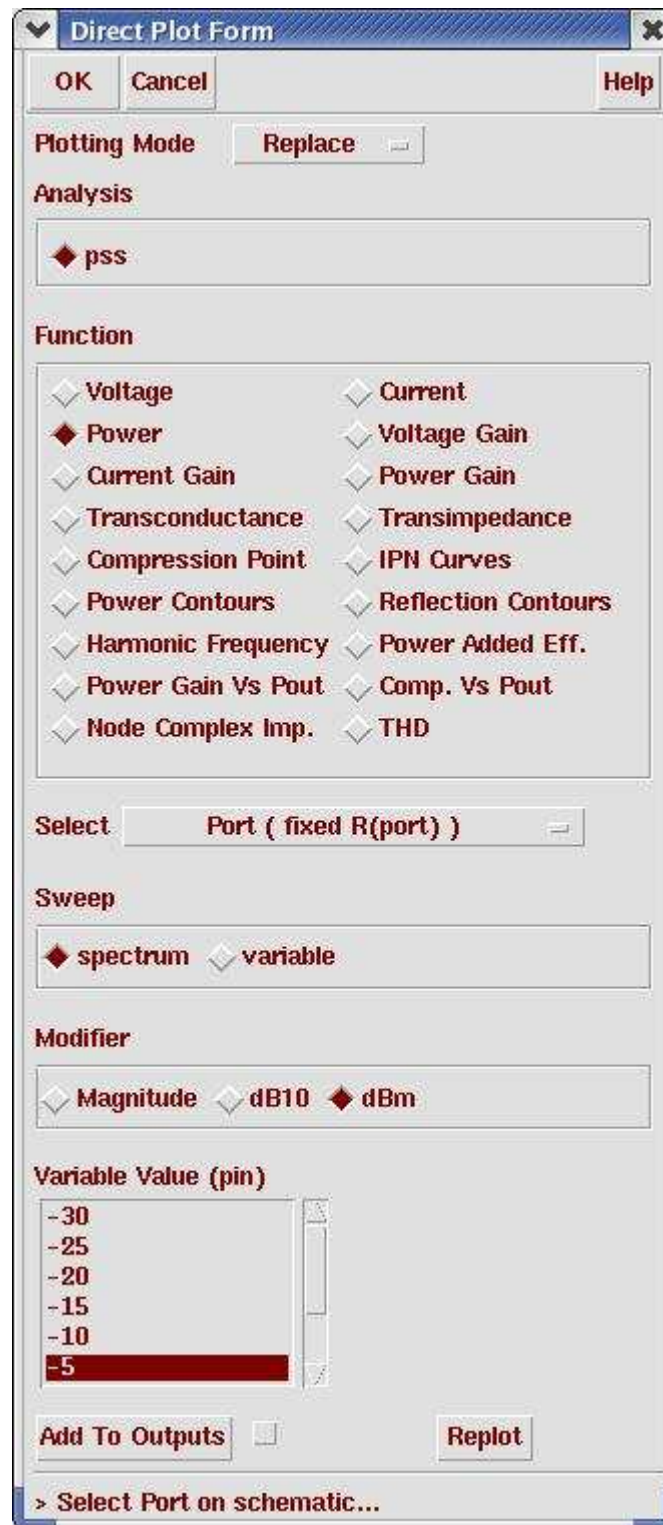
Action1-15: Select **Port2** on the schematic. The waveform window shows the output power VS input power..

PA Design Using SpectreRF



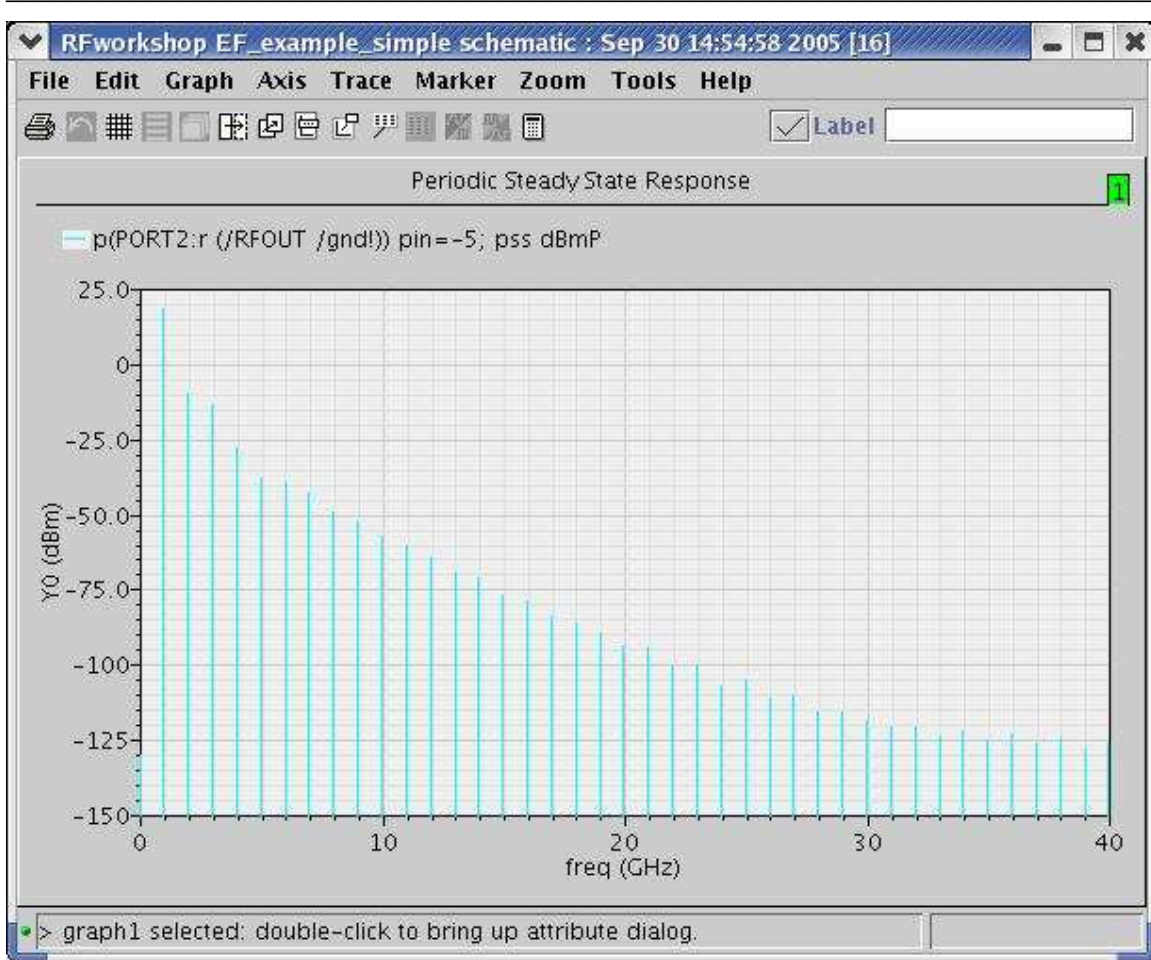
For the design example given, when the input power level is -5 dBm, the output power level is close to 20 dBm. Thus, -5 dBm is assumed to be the normal operating condition. All subsequent plots are based on this assumption.

Action1-16: In the Direct Plot form, change the Plot Mode to **replace** and configure the form as follows:



Action1-17: Click on Port2 to show the Power Spectrum.

PA Design Using SpectreRF

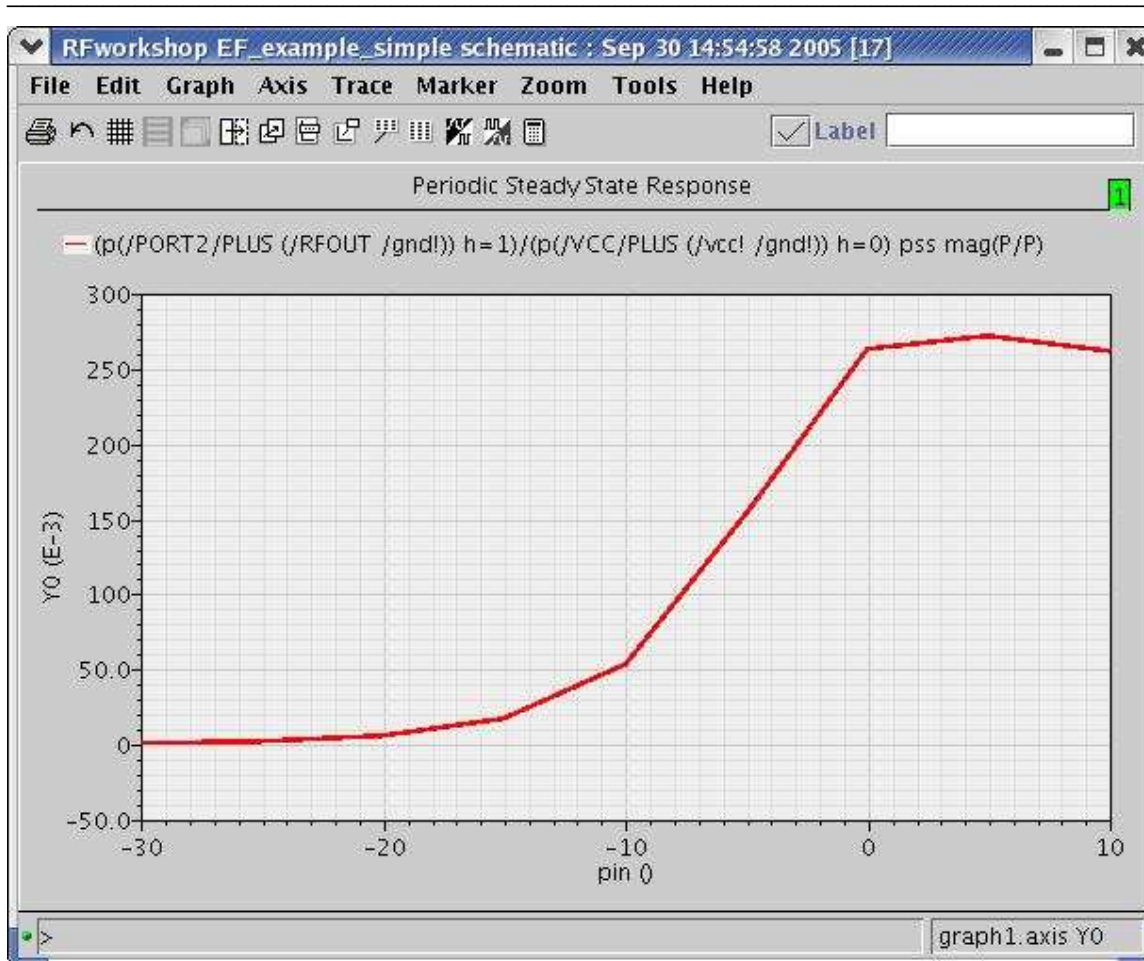


Action1-18: In the Direct Plot form, click Power Gain in the function field and configure the form as follows:



Action1-19: In the schematic, click the positive and negative terminals of Port2, and then click the positive and negative terminals of VCC.
The following plot shows the drain efficiency of the PA.

PA Design Using SpectreRF



Action1-20: In the waveform window, click the Add Subwindow icon.

SpectreRF provides a Power Added Efficiency (PAE) function. You only need to select the output terminal, input terminal and DC terminal in turn to plot the power added efficiency versus the input power level.

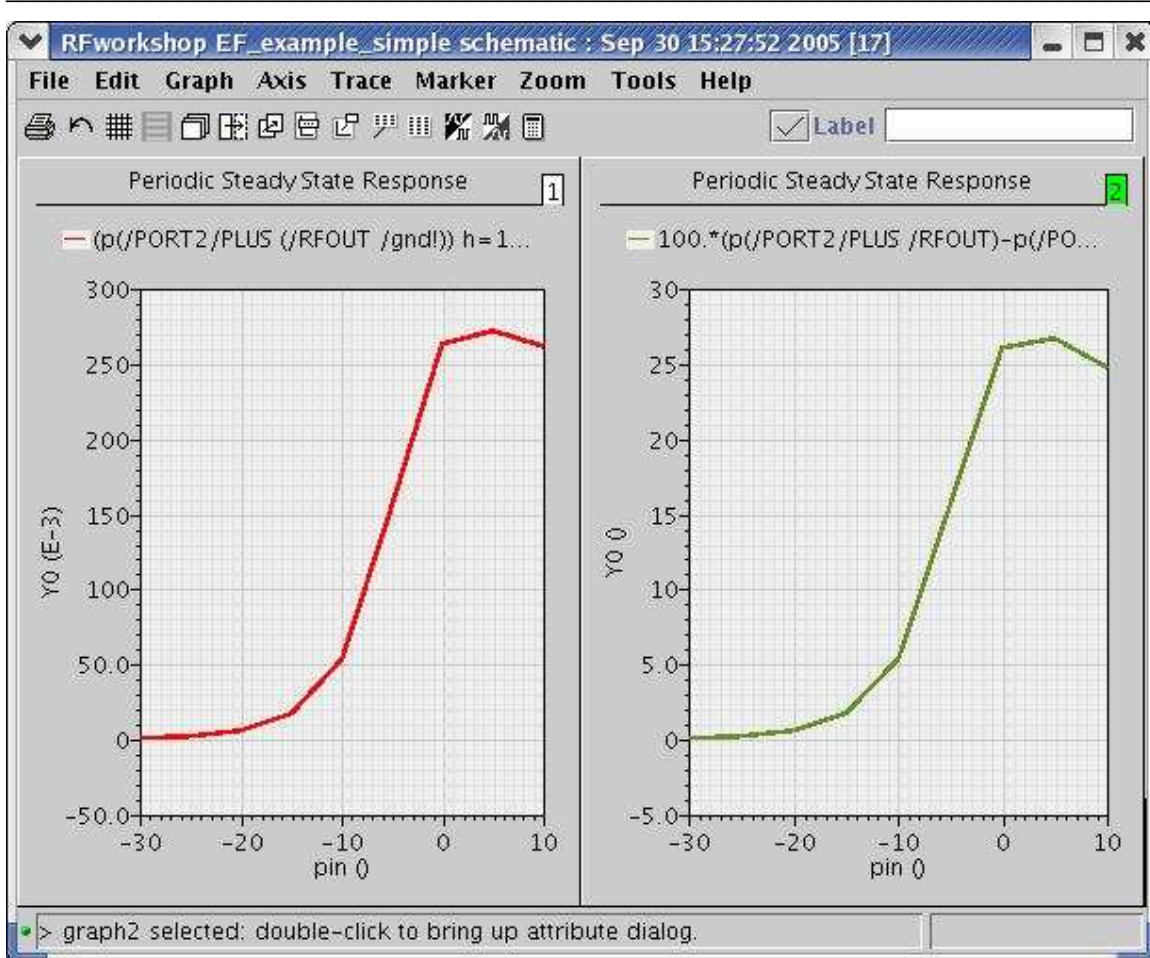
Action1-21: In the Direct Plot form, In the Direct Plot form, set the Plotting Mode to **Append**. Select **Power Added Eff.** in the Function field. Select the Output harmonic as 1GHz.



Action1-22: In the schematic, select the positive terminals of PORT2, PORT1 and VCC in turn..

The waveform window updates.

PA Design Using SpectreRF



Notice that for PA with high gain, the PAE is nearly equal to the drain efficiency. You will find that the efficiency of the PA around nominal operating conditions is only close to 20%.

Action1-23: Close the waveform window, the Direct Plot form and Virtuoso Analog Design Environment window.

Lab2: Linearity Measurement (Swept PSS with PAC)

1 dB compression point is defined as the input signal level that causes the small signal gain to drop by 1 dB. The suggested approach to measure 1 dB compression point is to set up a swept PSS analysis that sweeps input power level.

When the circuit is driven by two RF tones (f_{in} and f_{in2}), the third order intercept point is the intercept point of the first order fundamental power term (f_{in} , f_{in2}) and the third order intermodulation power term ($2 \times f_{in} - f_{in2}$, $2 \times f_{in2} - f_{in}$) expressed in some decibel form.

There are at least four ways to measure IIP3/OIP3 using SpectreRF:

1. PSS analysis with two large tones
2. QPSS analysis with one large tone and one moderate tone
3. Swept PSS and PAC analyses
4. Rapid IP3 using AC or PAC analysis

The recommended approach is to use method 3, swept PSS followed by PAC analyses, as this is faster and more accurate.

Action2-1: If not already open, Open the *schematic* view of the design *EF_example_simple* in the library *RFworkshop*

Action2-2: Select the **PORT1** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

Parameter	Value
Resistance	50 ohm
Port Number	1
DC voltage	(blank)
Source type	sine
Frequency name 1	RF
Frequency 1	fin
Amplitude 1 (dBm)	pin
PAC magnitude (dBm)	pin

- [Action2-3](#): Check and save the schematic.
- [Action2-4](#): From the EF_example_simple schematic, start the Virtuoso Analog Design Environment with the **Tools—Analog** Environment command.
- [Action2-5](#): You can choose **Session—Load State**, load state “**Lab2_IP3_PSSPAC**” and skip to [Action2-12](#) or ...
- [Action2-6](#): In Virtuoso Analog Design Environment, choose **Analyses—Choose...**
- [Action2-7](#): In the Choosing Analyses window, select the **pss** button in the **Analysis** field of the window.
- [Action2-8](#): Set up a swept PSS analysis as follows:

PA Design Using SpectreRF

Choosing Analyses – Virtuoso® Analog Design Environment

OK Cancel Defaults Apply Help

Engine ☐ Shooting ☐ Flexible Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
2	RF	fin	16	Large	PORT1

Large

Clear/Add Delete Update From Schematic

◆ Beat Frequency 16 Auto Calculate ■
◇ Beat Period

Output harmonics

Number of harmonics

Accuracy Defaults (empreset)

☐ conservative ■ moderate ☐ liberal

Additional Time for Stabilization (tstab)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Oscillator ☐

Sweep ■ Frequency Variable? ◆ no ◇ yes

Variable Variable Name pin Select Design Variable

Sweep Range

◆ Start-Stop Start -40 Stop 5
◇ Center-Span

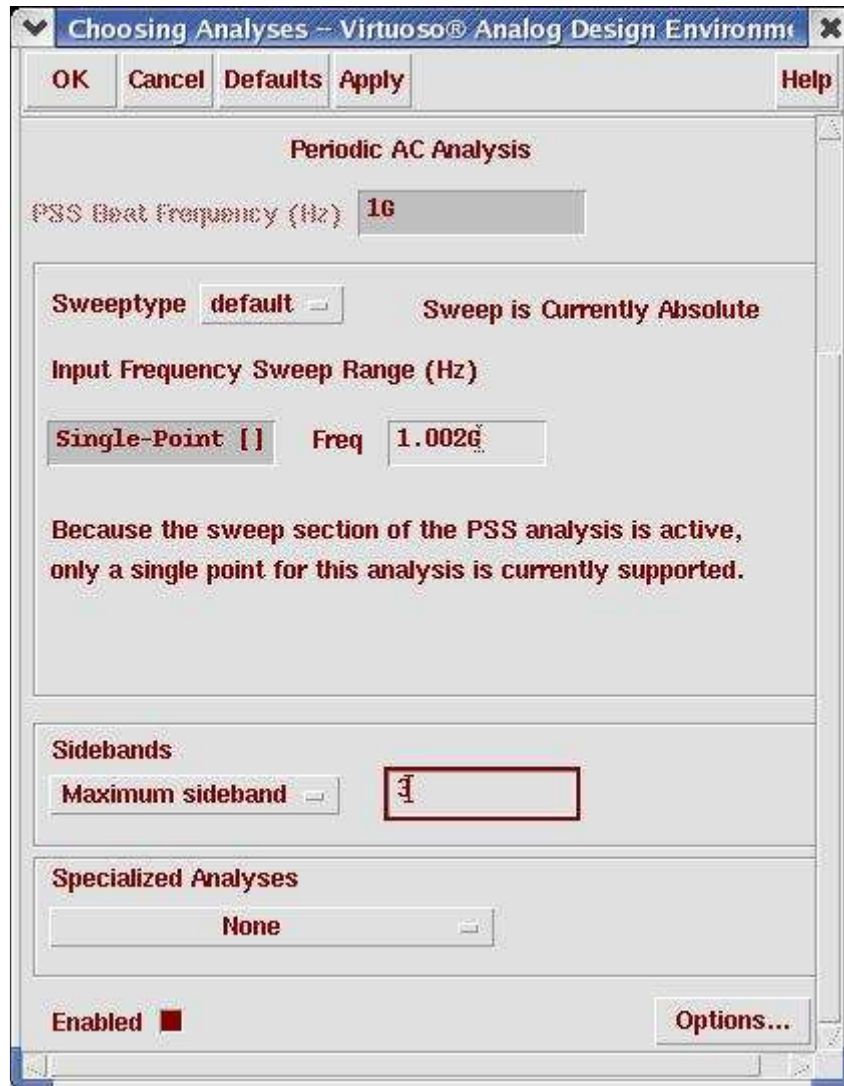
Sweep Type

◆ Linear Step Size 5
◇ Logarithmic Number of Steps

Add Specific Points ☐

Enabled ■ Options...

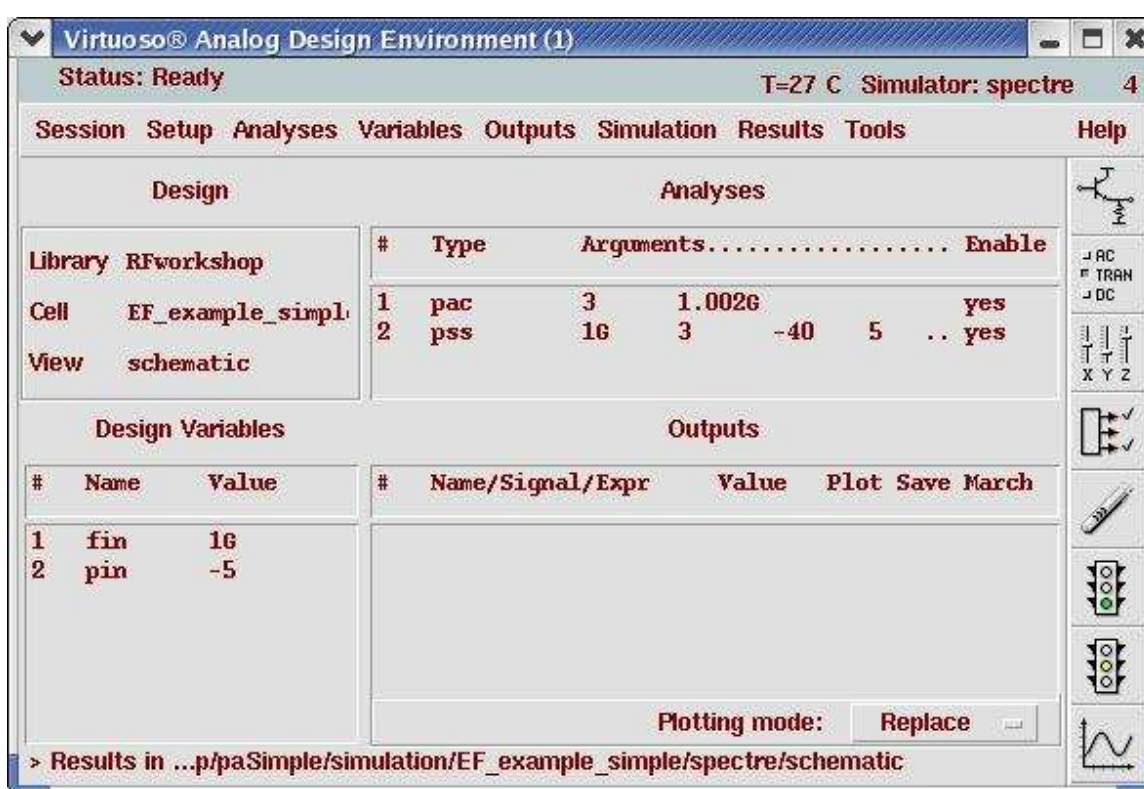
- Action2-9:** Make sure the **Enabled** button is active, and click **Apply** in the Choosing Analyses form.
- Action2-10:** In the Choosing Analyses window, select the **pac** button in the **Analysis** field of the window. Configure the form as follows:



- Action2-11:** Make sure the **Enabled** button is active, and click **OK** in the Choosing Analyses form

Your Virtuoso Analog Environment will look like this:

PA Design Using SpectreRF



Action2-12: In your Analog Design Environment, Choose **Simulation—Netlist and Run** or click the **Netlist and Run** icon to start the simulation.

When the simulation ends, plot the P1dB and IP3 curves.

Action2-13: In the Virtuoso Analog Design Environment, Choose **Results—Direct Plot—Main Form**.

Action2-14: In the Direct Plot Form, select the **pss** button and configure the form like this:

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

☒ pss ☐ pac

Function

☐ Voltage ☐ Current
☐ Power ☐ Voltage Gain
☐ Current Gain ☐ Power Gain
☐ Transconductance ☐ Transimpedance
☒ Compression Point ☐ IPN Curves
☐ Power Contours ☐ Reflection Contours
☐ Harmonic Frequency ☐ Power Added Eff.
☐ Power Gain Vs Pout ☐ Comp. Vs Pout
☐ Node Complex Imp. ☐ THD

Select Port (fixed R(port))

Format Output Power

Gain Compression (dB) 1

"pin" ranges from -40 to 5

Input Power Extrapolation Point (dBm)

(Defaults to -40)

Input Referred 1dB Compression

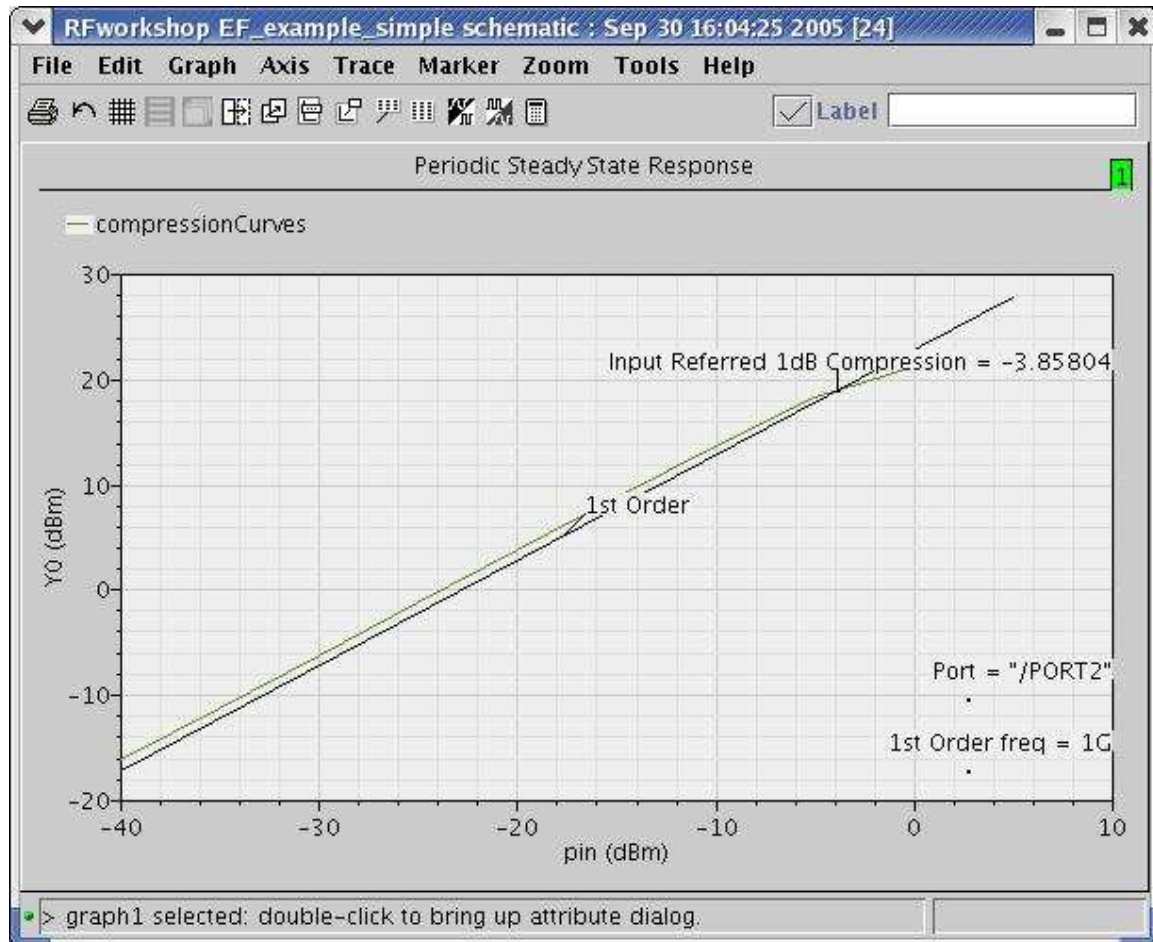
1st Order Harmonic

0	0
1	16
2	26
3	36
4	46
5	56

Add To Outputs ☐

> Select Port on schematic...

Action2-15: Click the **PORT2** to plot the 1 dB compression point.



Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

☐ pss ☒ pac

Function

☐ Voltage ☐ Voltage Gain
☐ Current ☒ IPN Curves

Select Port (fixed R(port))

Circuit Input Power ☐ Single Point
☒ Variable Sweep ("pin")

"pin" ranges from -40 to 5
 Input Power Extrapolation Point (dBm)
 (Defaults to -40)

Output Referred IP3 Order 3rd

3rd Order Harmonic		1st Order Harmonic	
-3	1.998G	-3	1.998G
-2	998M	-2	998M
-1	2M	-1	2M
0	1.002G	0	1.002G
1	2.002G	1	2.002G
2	3.002G	2	3.002G

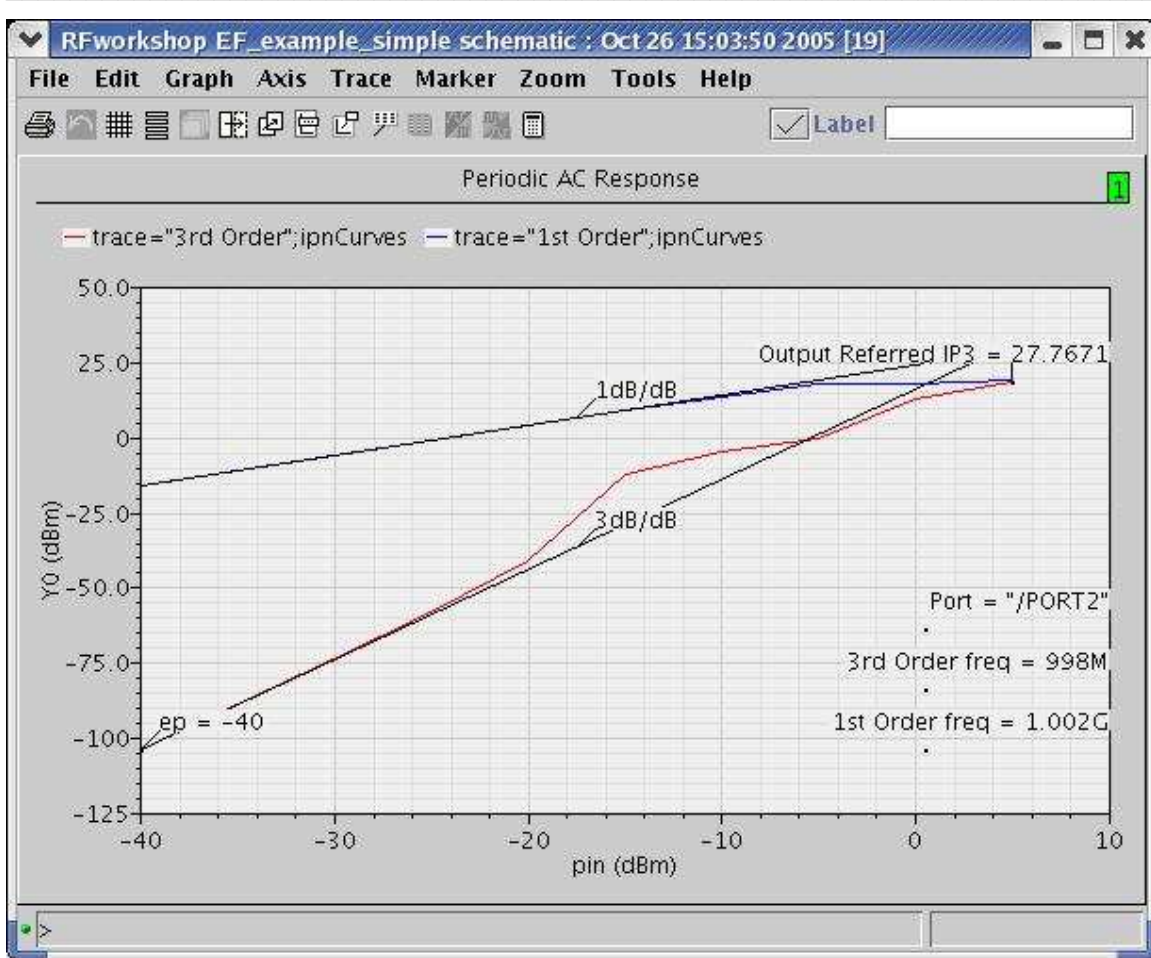
Add To Outputs ☐ Replot

freqaxis = absout

> Select Port on schematic...

Action2-17: Click the **PORT2** to plot the Output Referred IP3.

PA Design Using SpectreRF



Action2-18: Close the waveform window. Click **Cancel** on the Direct Plot form. Close the Virtuoso Analog Design Environment window.

Lab3: Stability and S-Parameter Measurements (PSS and PSP)

As pointed out by Gonzalez in [4], stability is guaranteed for the following conditions

$$Kf > 1, \Delta < 1$$

$$Kf > 1, B_{1f} = 1 + |S_{11}|^2 - |S_{22}|^2 - \Delta^2 > 0$$

To analyze stability for a PA, set up PSS and PSP analyses. The PSP analysis is a periodic small-signal analysis, so the S-parameter and VSWR results it generates apply only to the small signal. In some PA data sheets, the S-parameter and VSWR values specified are large signal characteristics. SpectreRF currently support large signal SP (LSSP) analysis. The LSSP analysis will show in Lab4.

Action3-1: If not already open, Open the *schematic* view of the design *EF_example_simple* in the library *RFworkshop*

Action3-2: Select the **PORT1** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

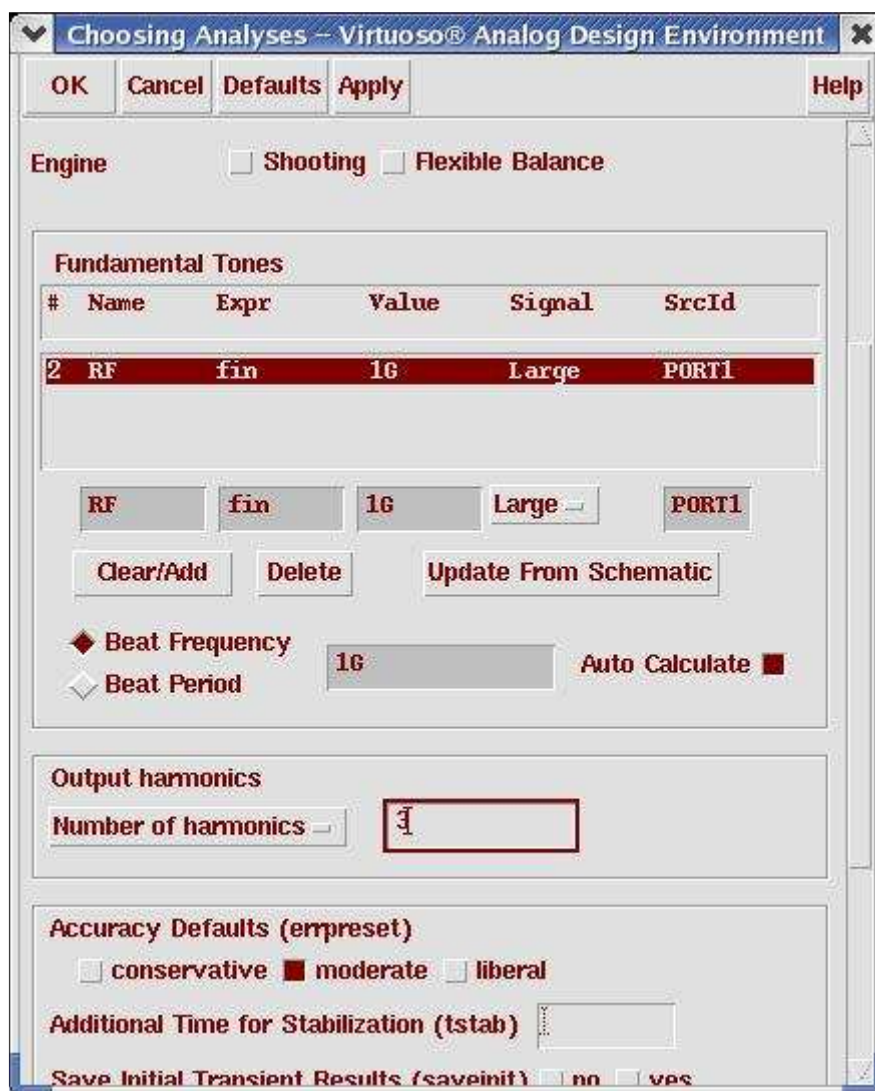
Parameter	Value
Resistance	50 ohm
Port Number	1
DC voltage	(blank)
Source type	sine
Frequency name 1	RF
Frequency 1	fin
Amplitude 1 (dBm)	pin

Action3-3: From the *EF_example_simple* schematic, start the Virtuoso Analog Design Environment with the **Tools—Analog** Environment command.

Action3-4: You can choose **Session—Load State**, load state “**Lab3_Stability_PSP**” and skip to [Action3-10](#) or ...

Action3-5: In Virtuoso Analog Design Environment, choose **Analyses—Choose...**

Action3-6: In the Choosing Analyses window, select the **pss** button in the **Analysis** field of the window and set up the form as follows:



Choosing Analyses – Virtuoso® Analog Design Environment

OK Cancel Defaults Apply Help

Engine ☐ Shooting ☐ Flexible Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
2	RF	fin	1G	Large	PORT1

RF fin 1G Large PORT1

Clear/Add Delete Update From Schematic

◆ Beat Frequency 1G Auto Calculate ■

◇ Beat Period

Output harmonics

Number of harmonics 1

Accuracy Defaults (errpreset)

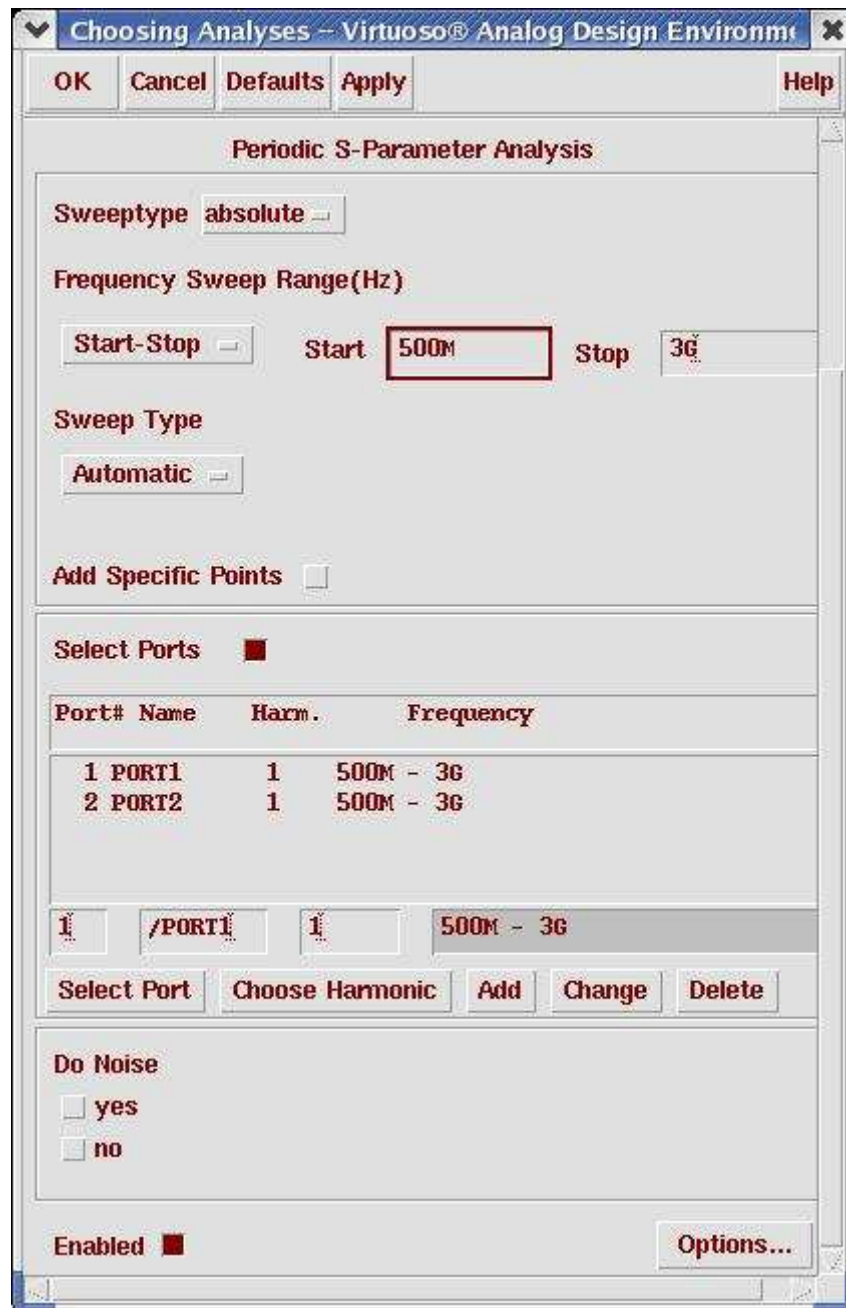
☐ conservative ☒ moderate ☐ liberal

Additional Time for Stabilization (tstab)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Action3-7: Make sure the **Enabled** button is active, and click **Apply** Choosing Analyses form.

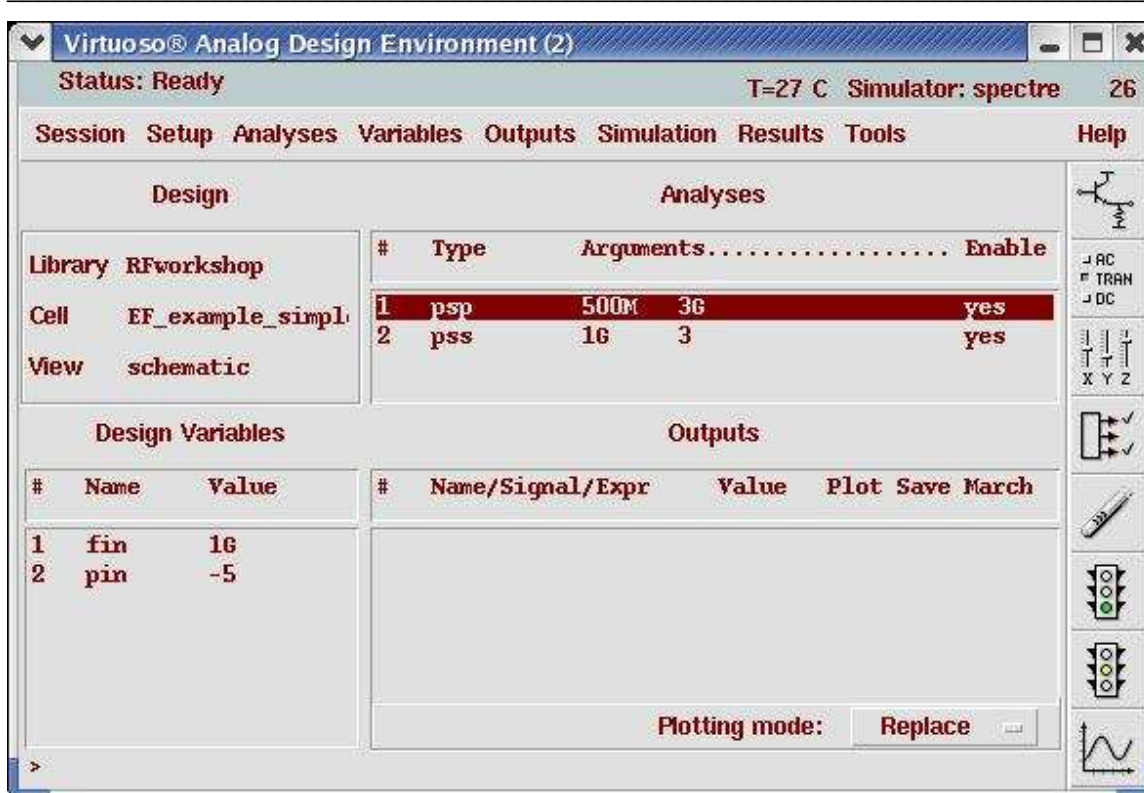
Action3-8: In the Choosing Analyses window, select the **psp** button in the **Analysis** field of the window and set up the form as follows:



Action3-9: Make sure the **Enabled** button is active, and click **OK**. The Choosing Analyses form.

Your Virtuoso Analog Environment will look like this:

PA Design Using SpectreRF



Action3-10: In your Analog Design Environment, Choose **Simulation—Netlist and Run** or click the **Netlist and Run** icon to start the simulation.

Action3-11: In the Virtuoso Analog Design Environment, Choose **Results—Direct Plot—Main Form**.

Action3-12: In the Direct Plot Form, select the **psp** button, Click **Kf** in the Function field. The form should look like this:

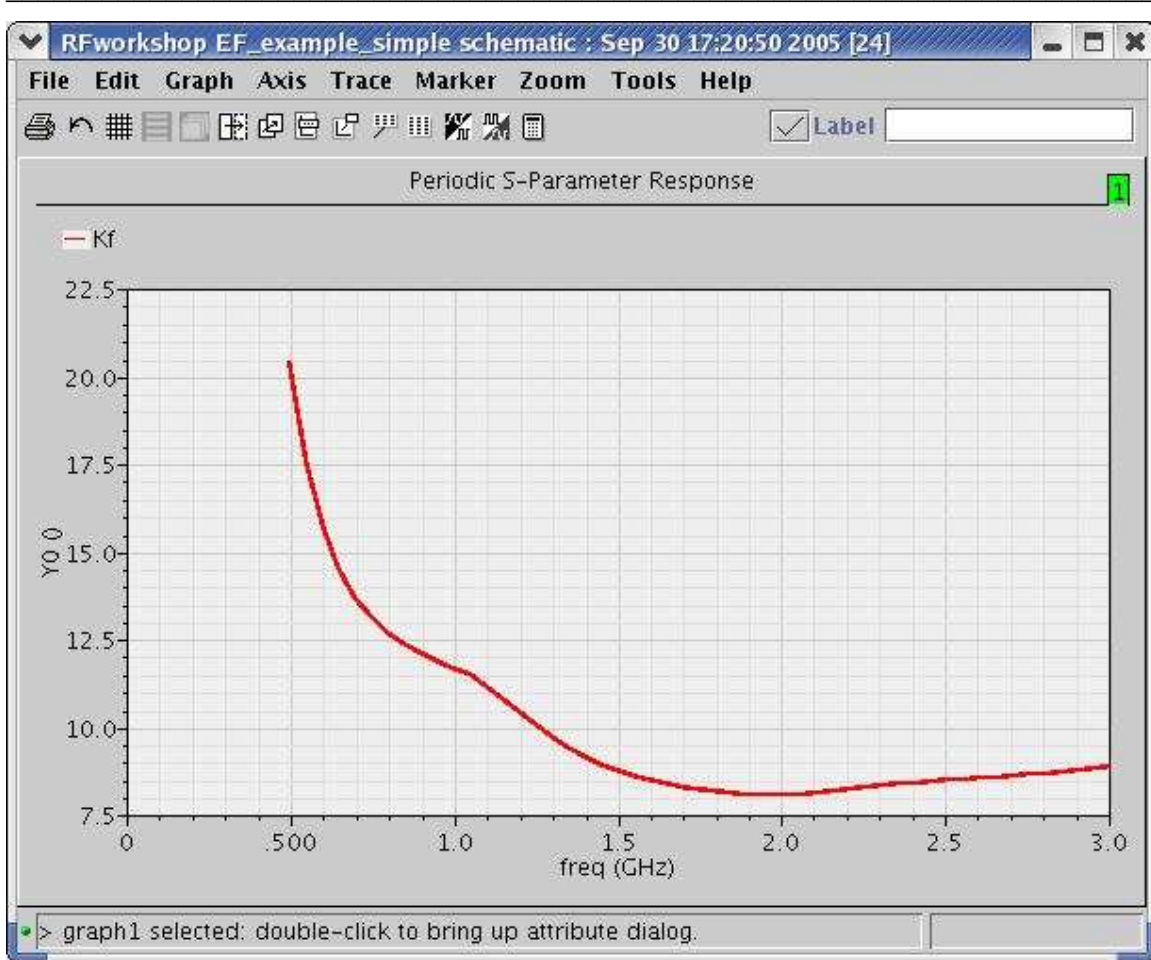
The image shows a 'Direct Plot Form' dialog box with the following sections:

- Buttons:** OK, Cancel, Help.
- Plotting Mode:** Replace.
- Analysis:** pss, **psp** (selected).
- Function:** A grid of checkboxes for various functions. The selected function is **Kf**.

☐ SP	☐ ZP	☐ YP	☐ HP
☐ GD	☐ VSWR	☐ NFmin	☐ Gmin
☐ Rn	☐ m	☐ NF	☒ Kf
☐ B1f	☐ GT	☐ GA	☐ GP
☐ Gmax	☐ Gmsg	☐ Gumx	☐ ZM
☐ NC	☐ GAC	☐ GPC	☐ LSB
☐ SSB	☐ F	☐ Fdsb	☐ Fieee
☐ Fmin	☐ GAIN	☐ IRN	☐ NFdsb
☐ NFieee			
- Description:** Stability Factor
- Buttons:** Add To Outputs, Plot.
- Footer:** > Press plot button on this form...

Action3-13: Click the **Plot** button. The following plot will show up.

PA Design Using SpectreRF



Action3-14: In the Direct Plot Form, select the **psp** button, Click **B1f** in the Function field. The form should look like this:

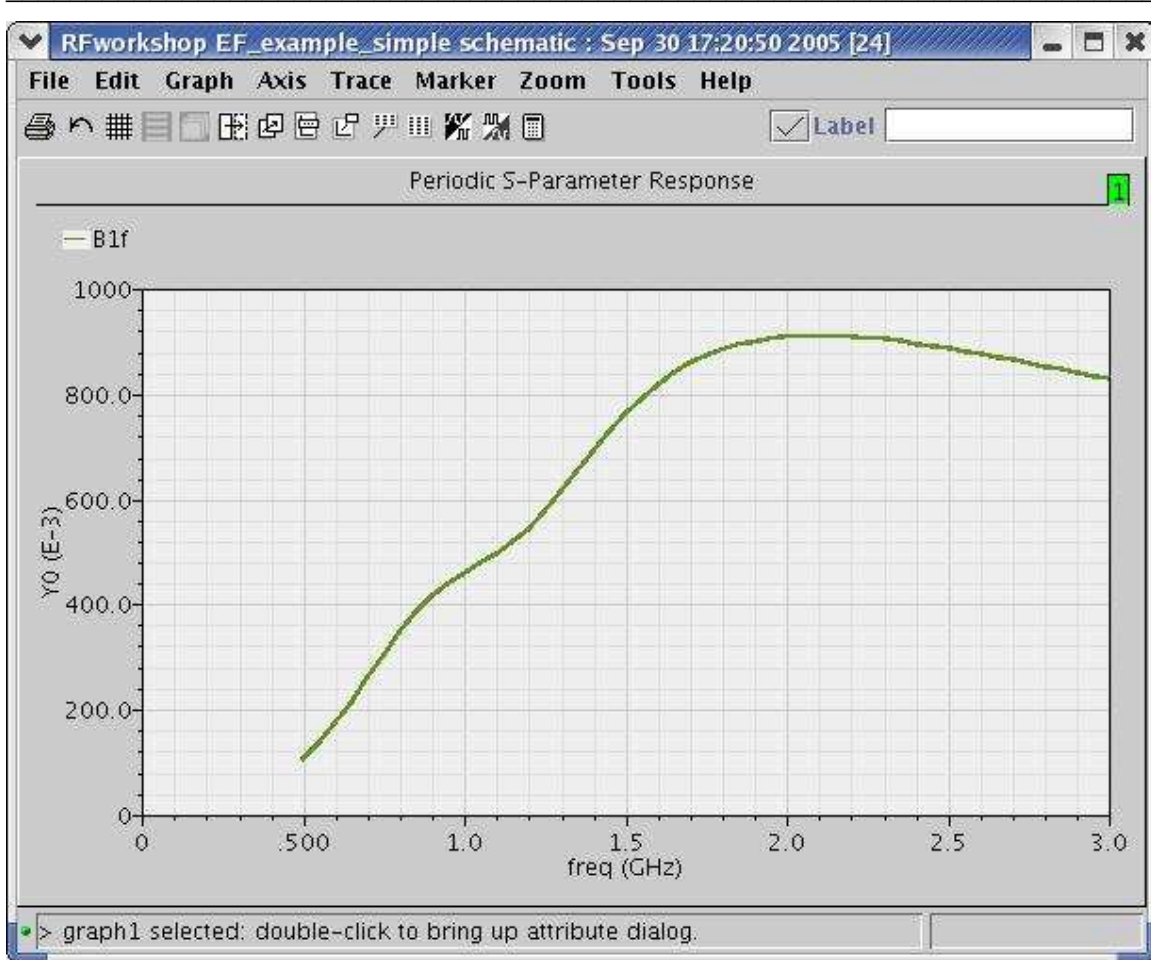
The image shows a 'Direct Plot Form' dialog box with the following sections:

- Buttons:** OK, Cancel, Help.
- Plotting Mode:** Replace.
- Analysis:** A list with 'pss' and 'psp', where 'psp' is selected with a diamond icon.
- Function:** A grid of 20 functions, each with a diamond icon. 'B1f' is selected.

SP	ZP	YP	HP
GD	VSWR	NFmin	Gmin
Rn	m	NF	Kf
B1f	GT	GA	GP
Gmax	Gmsg	Gumx	ZM
NC	GAC	GPC	LSB
SSB	F	Fdsb	Fieee
Fmin	GAIN	IRN	NFdsb
NFieee			
- Description:** Alternative Stability Factor.
- Buttons:** Add To Outputs (disabled), Plot.
- Footer:** > Press plot button on this form...

Action3-16: Click the **Plot** button. The following plot will show up.

PA Design Using SpectreRF



Action1-17: Close the waveform window.

Action1-18: In Direct Plot Form window, set **Plotting Mode** to **Append**. In the **Analysis** field, select **psp**. In the **Function** field, select **SP**. In the **Plot Type** field, select **Rectangular**. In the **Modifier** field, select **dB20**. The form should look like

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

☐ pss ☒ psp

Function

☒ SP	☐ ZP	☐ YP	☐ HP
☐ GD	☐ VSWR	☐ NFmin	☐ Gmin
☐ Rn	☐ m	☐ NF	☐ Kf
☐ B1f	☐ GT	☐ GA	☐ GP
☐ Gmax	☐ Gmsg	☐ Gumx	☐ ZM
☐ NC	☐ GAC	☐ GPC	☐ LSB
☐ SSB	☐ F	☐ Fdsb	☐ Fieee
☐ Fmin	☐ GAIN	☐ IRN	☐ NFdsb
☐ NFieee			

Description: S-Parameter

Plot Type

☒ Rectangular ☐ Z-Smith
☐ Y-Smith ☐ Polar

Modifier

☐ Magnitude ☐ Phase ☒ dB20
☐ Real ☐ Imaginary

S11 S12
S21 S22

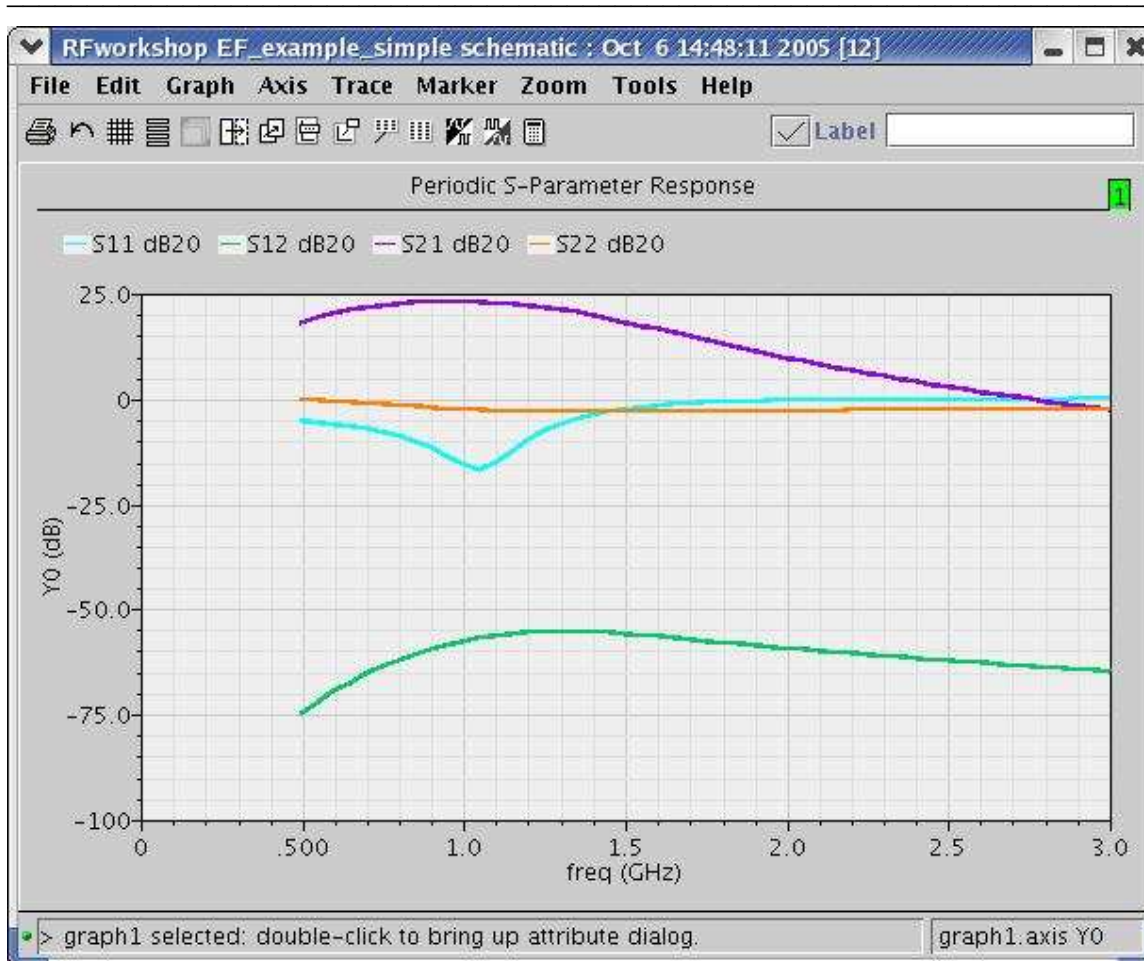
Port 1 active harmonic is 1
Port 2 active harmonic is 1

Add To Outputs ☐

> To plot, press Sij-button on this form...

Action1-19: Hit the S11, S12, S21 and S22 button.

PA Design Using SpectreRF



Action1-20: Close the waveform window.

Action1-21: In Direct Plot Form window, set **Plotting Mode** to **Append**. In the **Analysis** field, select **psp**. In the **Function** field, select **SP**. In the **Plot Type** field, select **Z-Smith**.

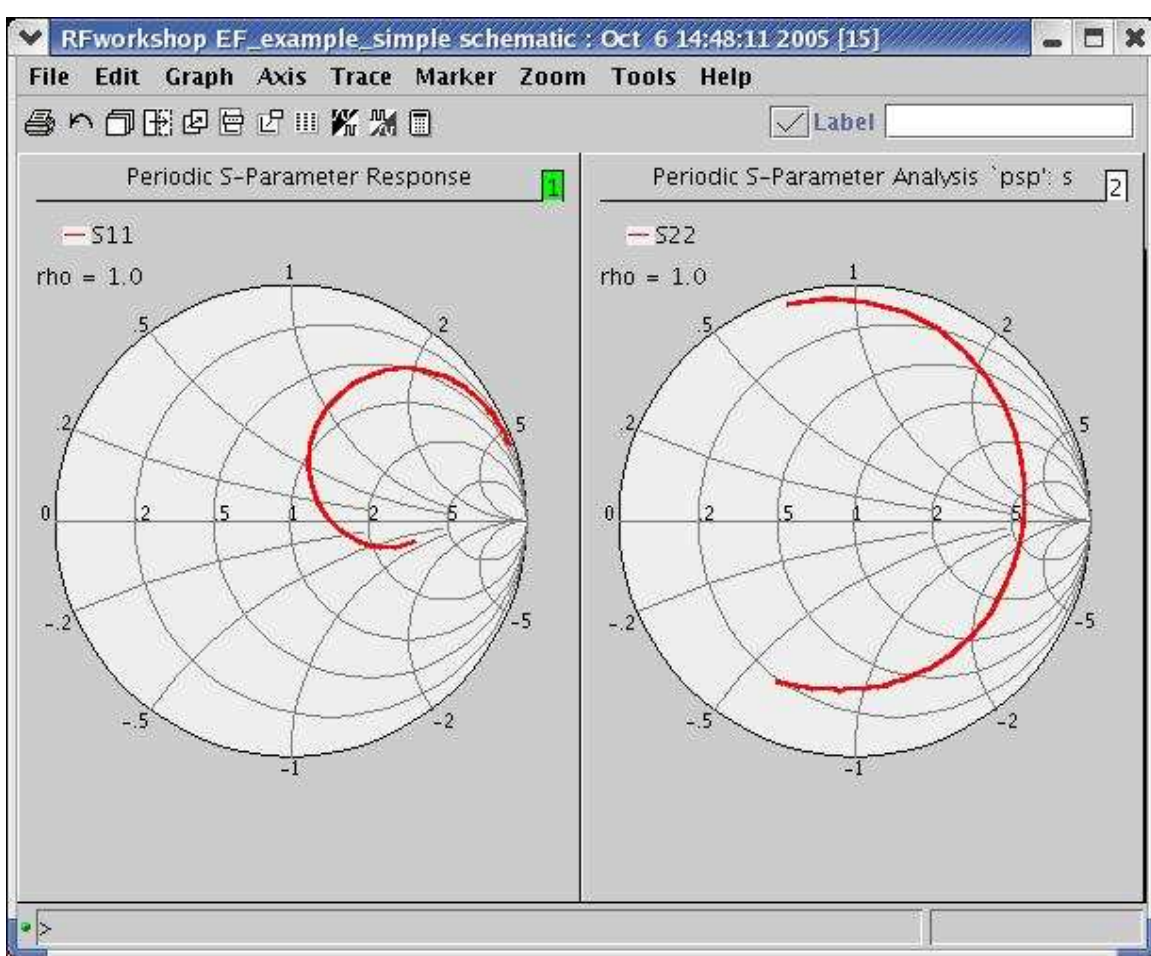
Action1-22: Click on **S11**. A waveform window appears.

Action1-23: In the waveform window, click on the **New Subwindow** button.

Action1-24: In the Direct Plot form, click on **S22**.

You plot the S11 and S22 in the Smith Chart.

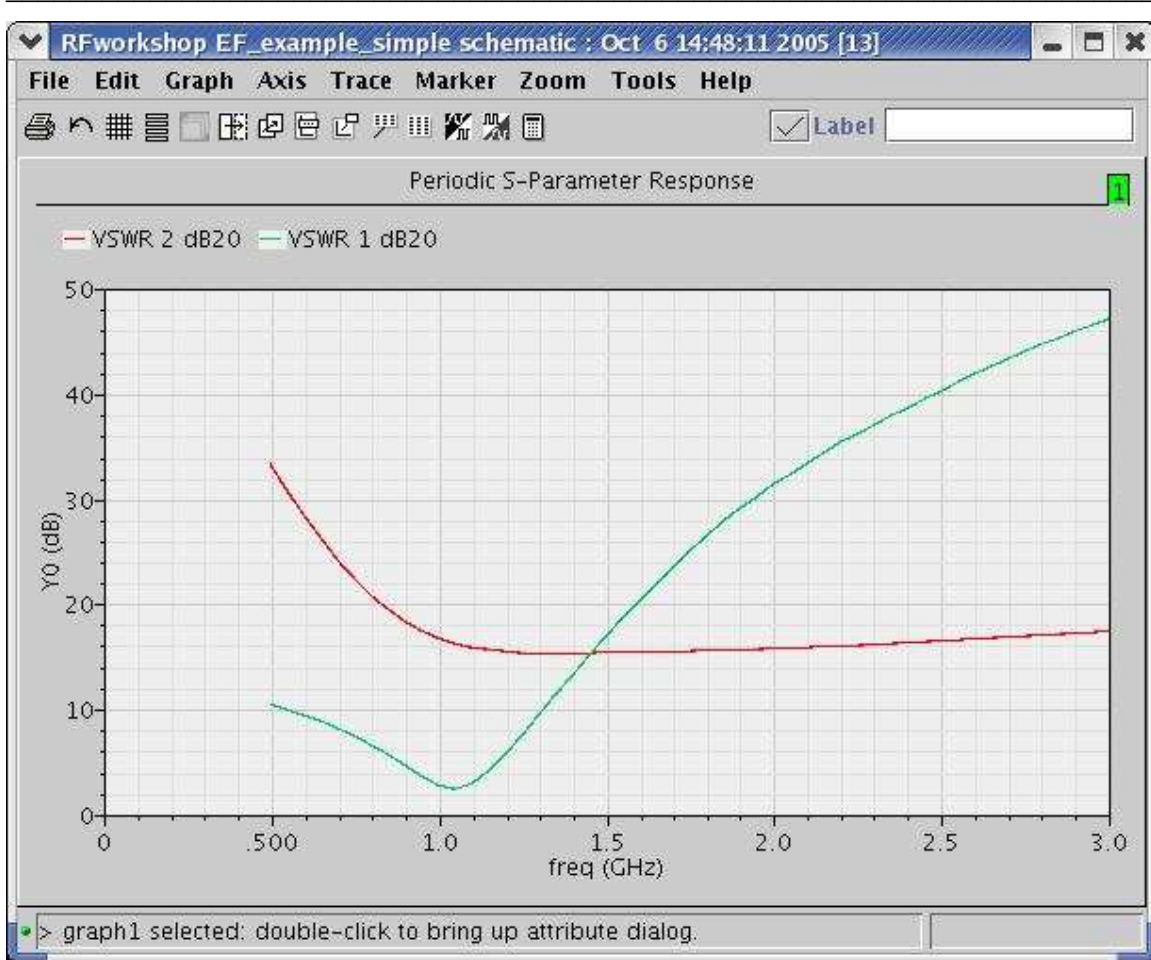
PA Design Using SpectreRF



Action1-25: In the Direct Plot Form window, in the **function** field, and choose **VSWR** (Voltage standing-wave ratio). In the **Modifier** field, select **dB20**. Press on **VSWR1**, then **VSWR2**.

You should get the following waveforms:

PA Design Using SpectreRF



Action3-26: Close the waveform window. Click **Cancel** on the Direct Plot form. Close the Virtuoso Analog Design Environment window.

Lab4. Large Signal S-Parameter Measurement (LSSP wizard)

The small-signal S-parameter characterization of an RF circuit is well established. However, for circuits with either large nonlinearity or frequency translations, small-signal S-parameters are not sufficient for design purposes. This is especially true for designs such as those that use power amplifiers and mixers.

As a natural extension of small-signal S-parameters, large-signal S-parameters can also be defined as the ratio of reflected (or transmitted) waves to incident waves. Since small-signal S-parameters are based on the simulation of a linearized circuit, small-signal S-parameters are independent of input power.

Large-signal S-parameters are based on large-signal steady state simulation techniques such as SpectreRF's PSS analysis with its shooting Newton method or harmonic balance simulators. Large-signal S-parameters are sensitive to input power levels.

Action4-1: If not already open, open the *schematic* view of the design *EF_example_LSSP* in the library *RFworkshop*

Action4-2: Select the **PORT1** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

Parameter	Value
Resistance	50 ohm
Port Number	1
DC voltage	(blank)
Source type	sine
Frequency name 1	RF
Frequency 1	fin
Amplitude 1 (dBm)	pin

Action4-3: Select the **PORT2** source. Use the **Edit—Properties—Objects** command to ensure that the port properties are set as described below:

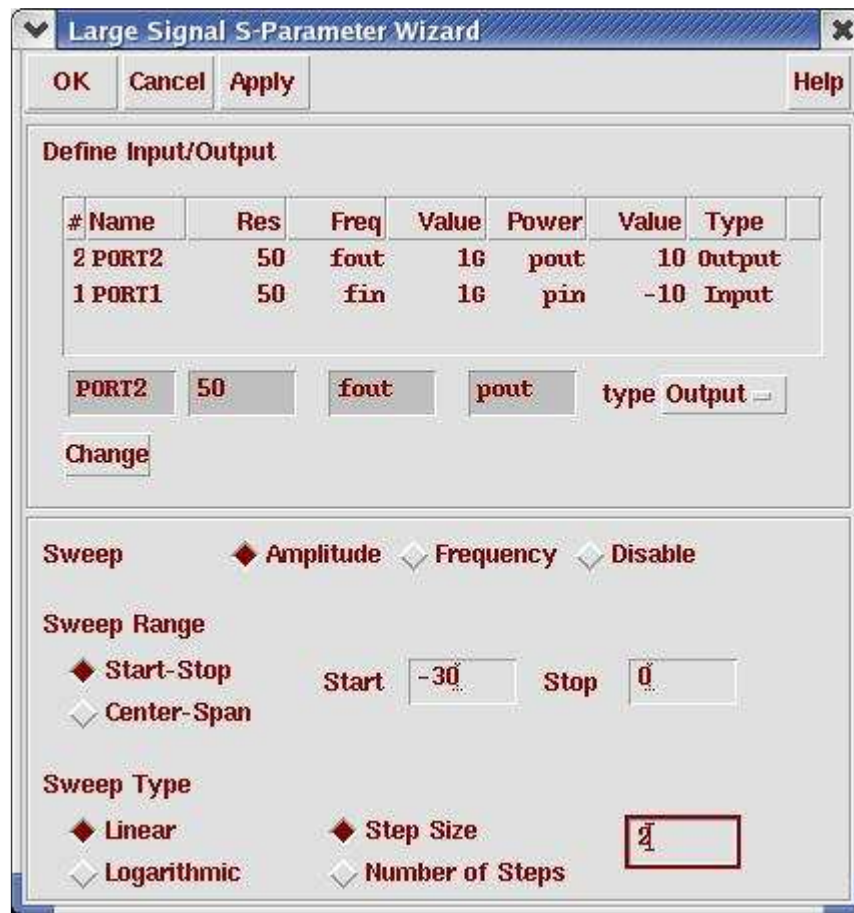
Parameter	Value
Resistance	50 ohm
Port Number	2
DC voltage	(blank)

PA Design Using SpectreRF

Source type	sine
Frequency name 1	RFout
Frequency 1	fout
Amplitude 1 (dBm)	pout

Make sure you are using PORT. SpectreRF currently only support PORT for LSSP simulation.

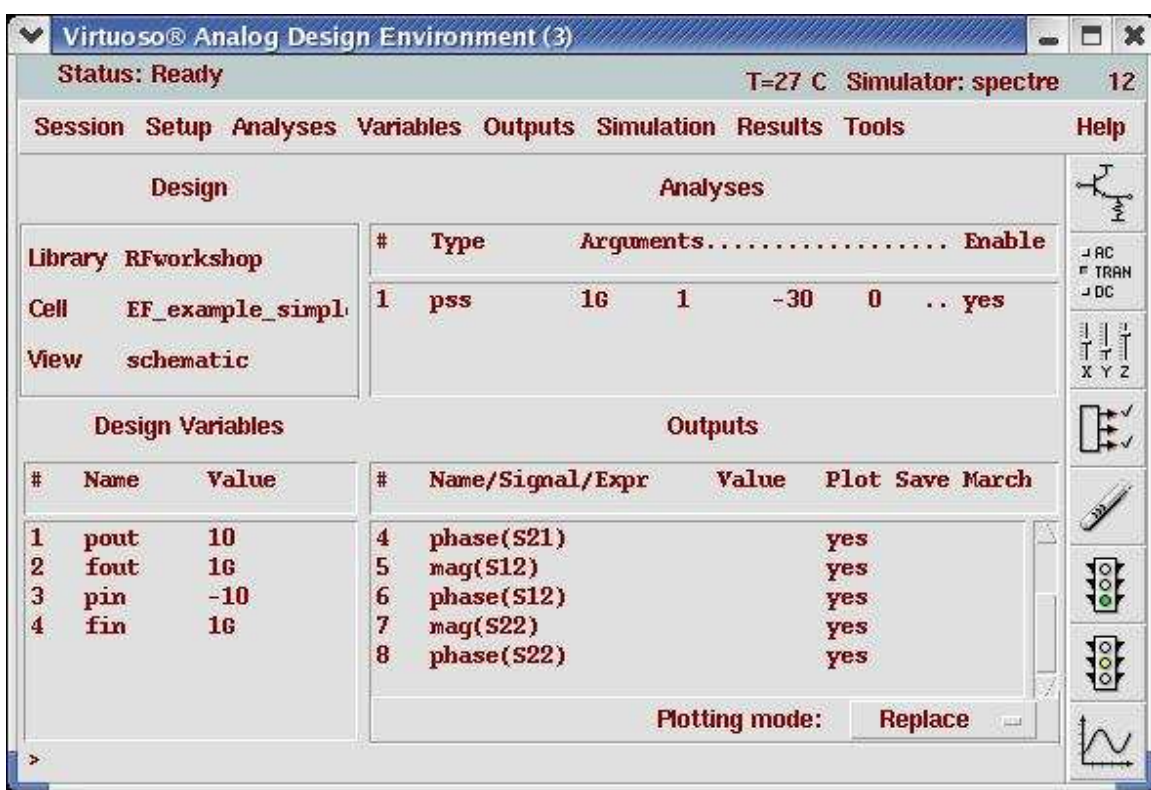
- Action4-4:** Check and save the schematic.
- Action4-5:** From the EF_example_simple schematic, start the Virtuoso Analog Design Environment with the **Tools—Analog** Environment command.
- Action4-6:** In Vituoso Analog Design Environment, choose **Tools—RF---Wizards--LSSP**
- Action4-7:** In the Large Signal S-Parameter Wizard window, select **Port1** in the field of **Define Input/Output**.
- Action4-8:** Change **Type** to **Input**.
- Action4-9:** Select **Port2** and change type to **Output**.
- Action4-10:** In the Large Signal S-Parameter Wizard window, choose Amplitude in **Sweep** field.
- Action4-11:** Configure the forms as follows:



Action4-12: In the Large Signal S-Parameter Wizard window, click **OK** to close the window.

Your Virtuoso Analog Environment will look like this:

PA Design Using SpectreRF

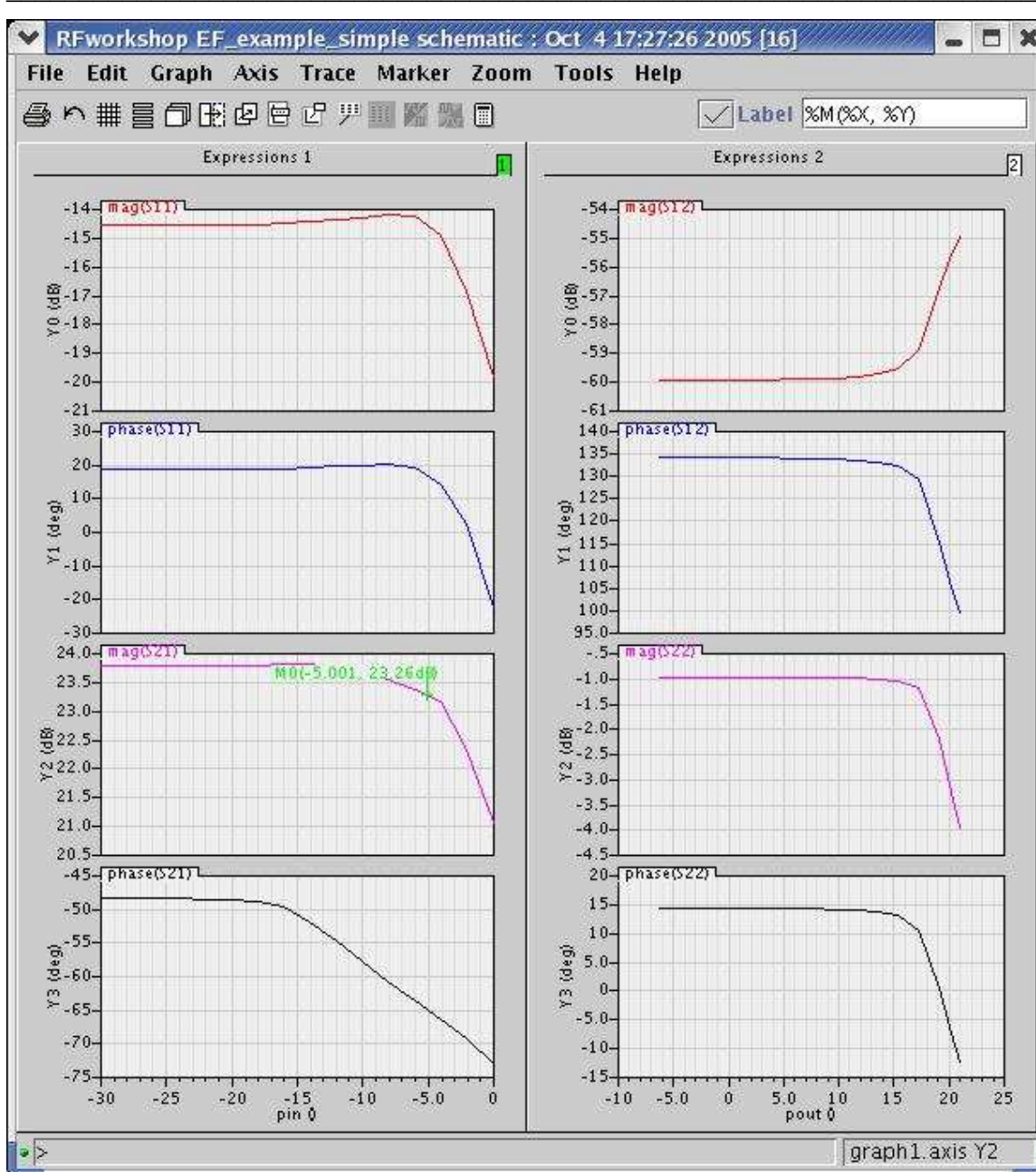


Action4-13: In your Analog Design Environment, Choose **Simulation—Netlist and Run** or click the **Netlist and Run** icon to start the simulation.

After the simulation ends, the waveform window appears.

Action4-14: In the waveform window, place a marker in curve mag(S21) at Pin=-5 dBm by choosing **Marker—Place—Trace marker**. It shows that S21=23.26dB at Pin=-5 dBm.

PA Design Using SpectreRF



Action4-15: In Virtuoso Analog Design Environment window, choose **Variable—Edit**. Editing Design Variable window appears.

Action4-16: In the Editing Design Variable window, click on **pin -10**, change its value to **-5**. Click on **Change**.

Action4-17: In the Editing Design Variable window, click on **pout 10**, change its value to **18.26**. Click on **Change**.

The PA output will be $-5 + 23.26 = 18.26$ dBm when $\text{pin} = 5$ dBm.

Action4-18: Click on OK in the Editing Design Variable window.

Action4-19: In Virtuoso Analog Design Environment, choose **Tools—RF---Wizards--LSSP**

Action4-20: Configure the Large Signal S-Parameter Wizard form as follows:

#	Name	Res	Freq	Value	Power	Value	Type
2	PORT2	50	fout	1G	pout	18.26	Output
1	PORT1	50	fin	1G	pin	-5	Input

PORT2 50 fout pout type Output

Change

Sweep ☐ Amplitude ☒ Frequency ☐ Disable

Sweep Range

☒ Start-Stop Start 500M Stop 3G

☐ Center-Span

Sweep Type

☒ Linear ☐ Logarithmic

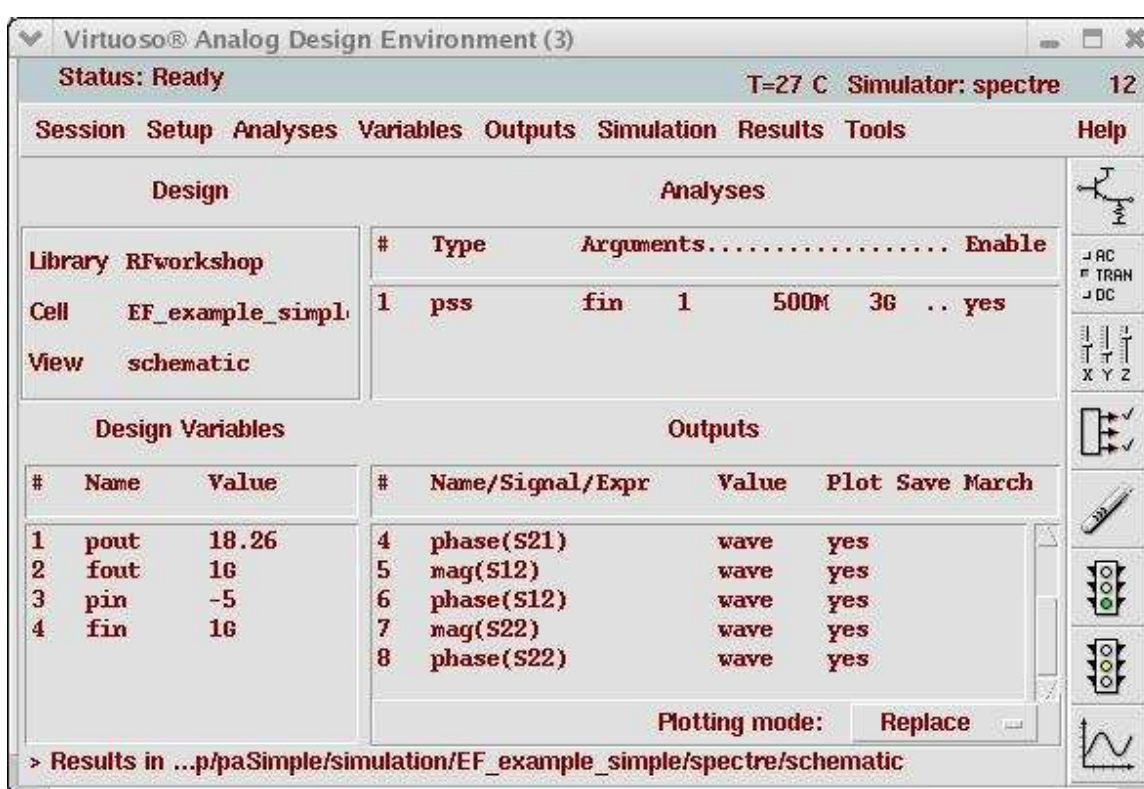
☒ Step Size 100M

☐ Number of Steps

Action4-21: In the Large Signal S-Parameter Wizard window, click **OK** to close the window.

Your Virtuoso Analog Environment will look like this:

PA Design Using SpectreRF

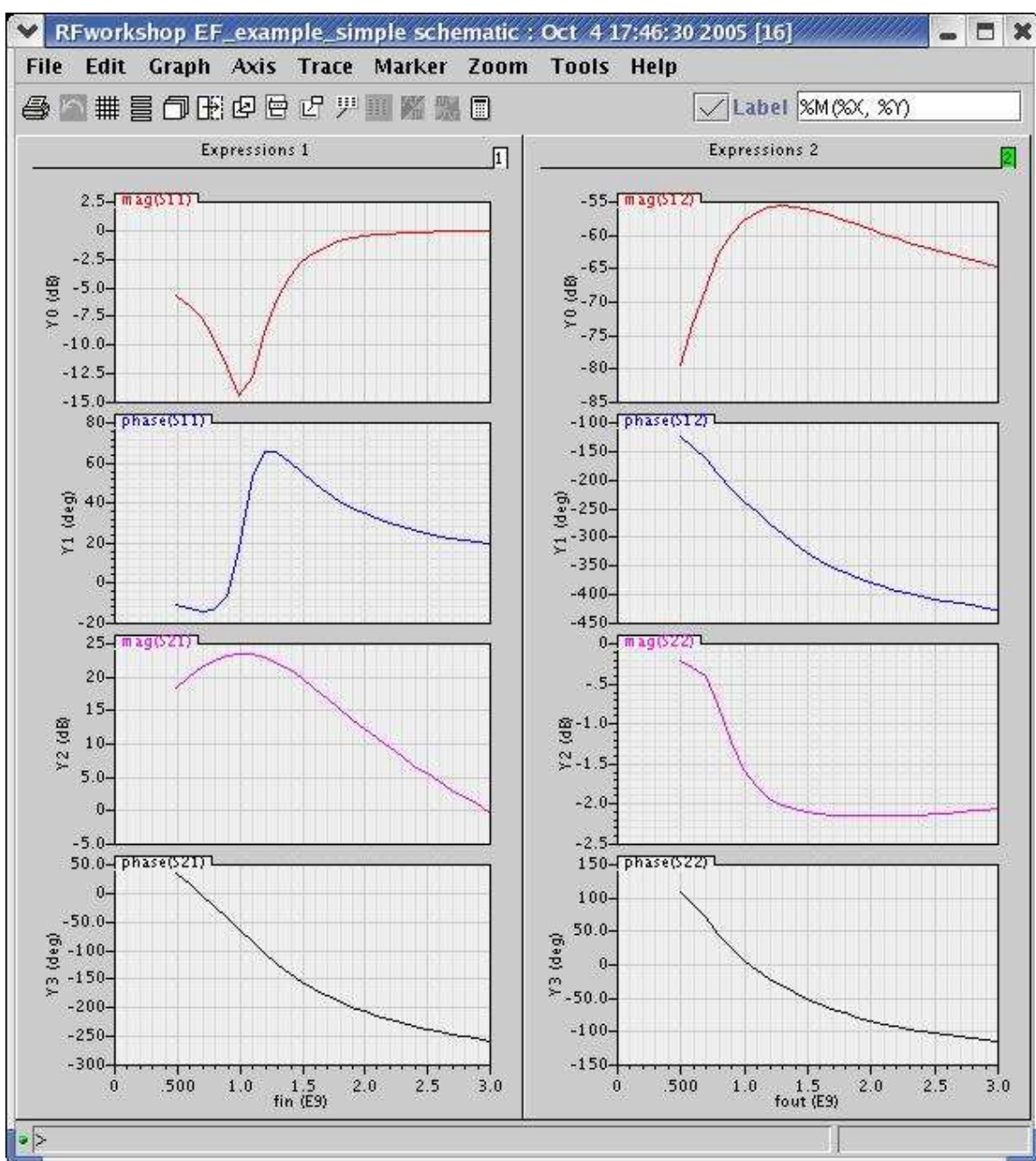


Action4-22: In your Analog Design Environment, Choose **Simulation—Netlist and Run** or click the **Netlist and Run** icon to start the simulation

After the simulation ends, the waveform window appears.

Note: You may want to change the graph to strip mode to get individual graph I each subwindow if the strip mode is not set by default.

PA Design Using SpectreRF



Action4-23: Close the waveform window. Click **Cancel** on the Direct Plot form. Close the Virtuoso Analog Design Environment window. Close the *EF_example_LSSP* schematic.

Lab5: Load-Pull Measurements (Swept PSS)

A load pull analysis is a systematic way to measure large signal impedance matching. In a load pull analysis, the output reflection coefficients are swept; SpectreRF measures the output power and plots it as a function of the complex load as seen by the transistor. Since the complex load requires two axes, the results are plotted as constant power contours on a Smith chart. The contours show how the output power increases as the load impedance reaches its optimum value, Z_{opt} .

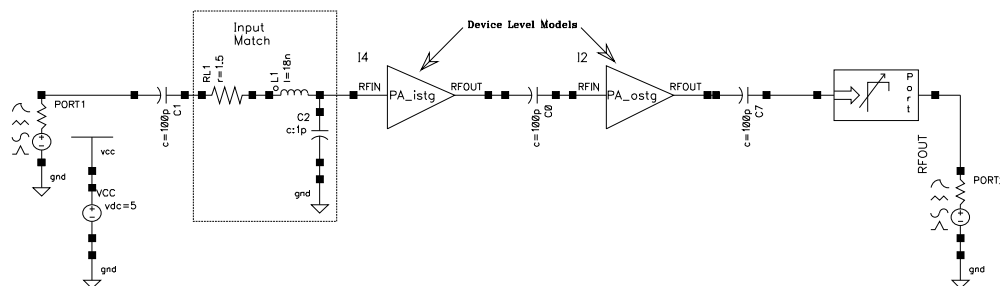
Keep in mind that you are sweeping output reflection coefficients by changing a linear load. The large signal output reflection coefficients computed in this manner equal the small-signal, or incrementally computed, load reflection coefficients. However, for input reflection coefficients this is no longer true. Actually, you are computing the large signal reflection coefficients at the fundamental frequency.

You might not always be able to achieve the optimal output power due to other design goals such as stability concerns for instance. Those goals are generally posed as constraints in the reflection coefficients. SpectreRF allows you to overlay the reflection coefficients on top of the constant power contours and make your design choices.

However, a constant power contour does not equal a constant power gain contour. You should plot the input power contours both to verify that the PA's input impedance does not change significantly as the load impedance changes and to ensure that you have achieved a reasonable power gain.

Action5-1: Open the *schematic* view of the design *EF_example_loadpull* in the library *RFworkshop*

The following figure shows the modified *EF_sxample-simple* schematic for frequency pull calculations.



The input port in the above testbench has the following parameters:

Parameter	Value
Resistance	50 ohm
Port Number	1
DC voltage	(blank)
Source type	sine
Frequency name 1	RF
Frequency 1	fin
Amplitude 1 (dBm)	pin

The output port has the setup as below:

Parameter	Value
Resistance	50 ohm
Port Number	2
DC voltage	(blank)
Source type	dc

An instance of a PortAdaptor is connected to the load. The PortAdaptor is set to have the following properties:

- Frequency = 1.115 G;
- Phase of Gamma = theta;
- Mag of Gamma = 0.2512
- Reference Resistance = 10K (this value must equal to the load).

Action5-2: From the *EF_sxample-loadpull* schematic, start the Virtuoso Analog Design Environment with the **Tools—Analog** Environment command.

Action5-3: You can choose **Session—Load State**, load state “**Lab5_LoadPull_PSS**” and skip to [Action5-10](#) or ...

Action5-4: In Virtuoso Analog Design Environment, choose **Analyses—Choose...**

Action5-5: In the Choosing Analyses window, select the **pss** button in the **Analysis** field of the window.

Action5-6: Set up a swept PSS analysis with the theta parameter varying from 0 to 359 degrees. Set Beat Frequency = 1G; Number of Harmonics = 10; errpreset = moderate; enable the Sweep button; enter theta as Variable

PA Design Using SpectreRF

Name; set the Sweep Range Start = 0 and Stop = 359; set Sweep Type = linear; and Number of Steps = 10.

Your PSS analysis window should look like...

Choosing Analyses – Virtuoso® Analog Design Environment

OK Cancel Defaults Apply Help

Engine ☐ Shooting ☐ Flexible Balance

Fundamental Tones

#	Name	Expr	Value	Signal	SrcId
2	RF	fin	1G	Large	PORT1

Large

Clear/Add Delete Update From Schematic

◆ Beat Frequency 1G Auto Calculate ■

◇ Beat Period

Output harmonics

Number of harmonics 10

Accuracy Defaults (errpreset)

☐ conservative ■ moderate ☐ liberal

Additional Time for Stabilization (tstab)

Save Initial Transient Results (saveinit) ☐ no ☐ yes

Oscillator ☐

Sweep ■ Frequency Variable? ◆ no ◇ yes

Variable Variable Name theta

Select Design Variable

Sweep Range

◆ Start-Stop Start 0 Stop 359

◇ Center-Span

Sweep Type

◆ Linear ◇ Step Size 10

◇ Logarithmic ◆ Number of Steps 10

Add Specific Points ☐

Enabled ■ Options...

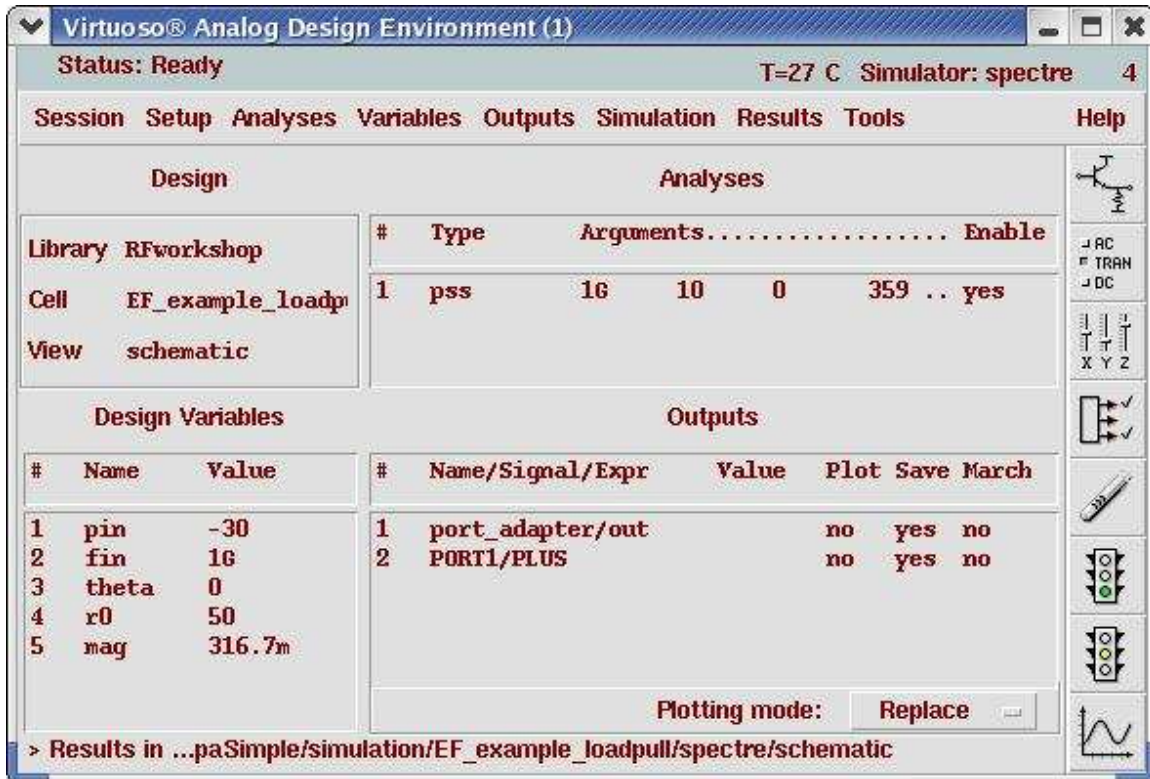
PA Design Using SpectreRF

Action5-7: Make sure the **Enabled** button is active, and click **OK** in the Choosing Analyses form.

Action5-8: In the Virtuosos Analog Design Environment window, Choose **Outputs—To be Saved—Select on Schematic**

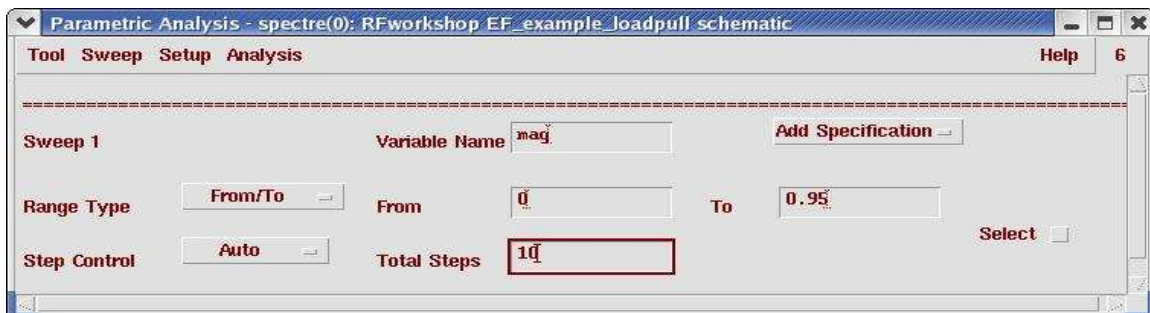
Action5-9: In the schematic, select the input terminals of PORT1 and portAdapter. Press ESC key to escape the selection process.

Your Virtuoso Analog Environment will look like this:



Action5-10: In your Virtuoso Analog Design Environment window, click **Tools—Parametric Analysis...**

The Parametric analysis form appears. Configure the form as below:



- Action5-11:** In your Parametric Analysis form, choose **Analysis—Start...**
- Action5-12:** In the Virtuoso Analog Design Environment, select **Session—Options**, change the **Waveform Tool** to **AWD**.
- Action5-13:** After the simulation has run, in the Virtuoso Analog Design Environment, Choose **Results—Direct Plot—Main Form**.
- Action5-14:** In the Direct Plot Form, select the **pss** button, choose the **Power Contours** function. Make sure **Select** is toggled to **Single Power/Refl Terminal**, select **fundamental (harmonic 1)** as the output harmonic. . The form should look like this:

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

pss

Function

- ☐ Voltage
- ☐ Current
- ☐ Power
- ☐ Voltage Gain
- ☐ Current Gain
- ☐ Power Gain
- ☐ Transconductance
- ☐ Transimpedance
- ☐ Compression Point
- ☐ IPN Curves
- ☒ Power Contours
- ☐ Reflection Contours
- ☐ Harmonic Frequency
- ☐ Power Added Eff.
- ☐ Power Gain Vs Pout
- ☐ Comp. Vs Pout
- ☐ Node Complex Imp.
- ☐ THD

Select Single Power/Refl Terminal

Power Modifier ☒ Magnitude ☐ dB10 ☐ dBm

Maximum Power

Minimum Power

Reference Resistance 50.0

Number of Contours 9

Close Contours

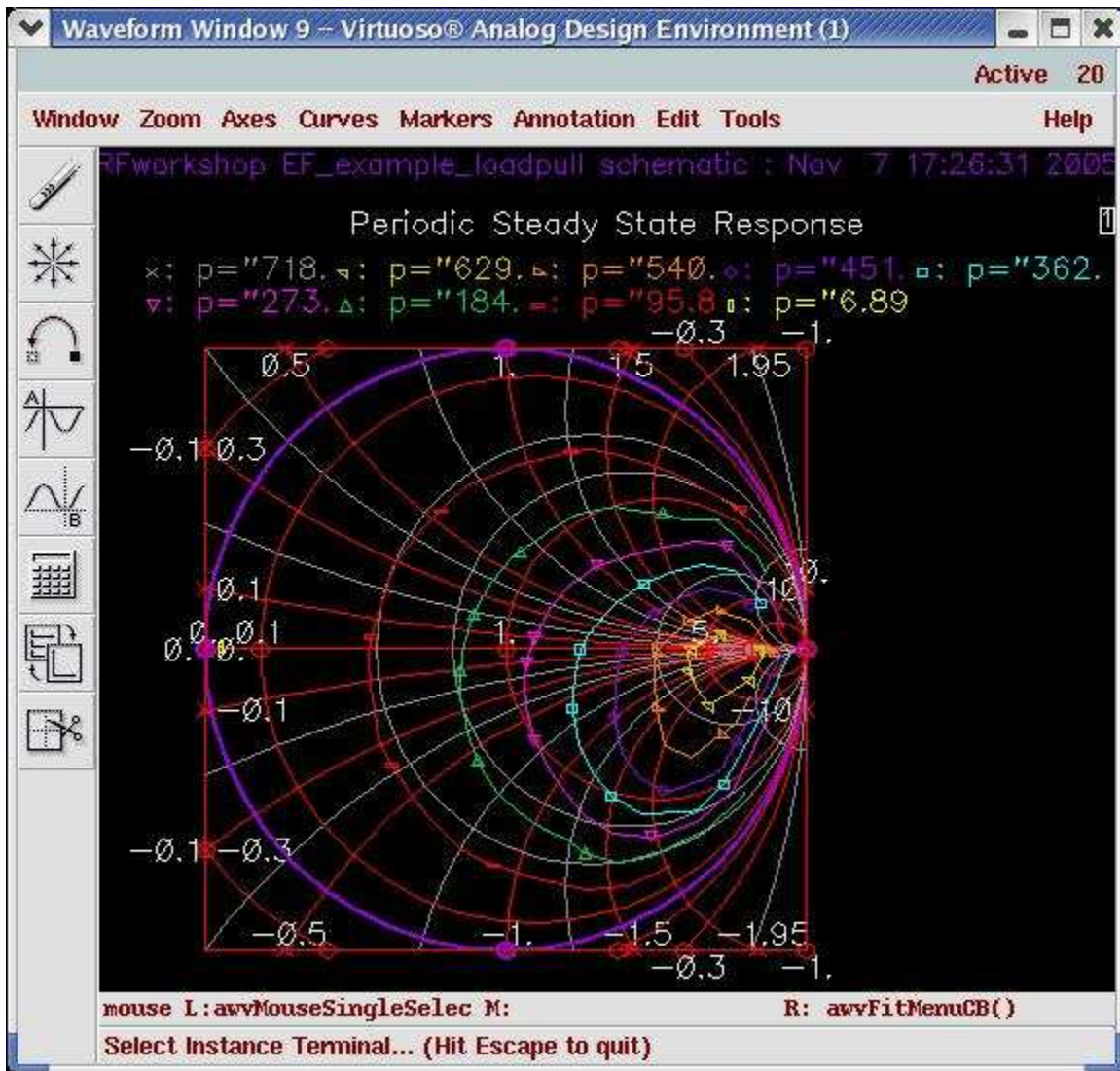
Output Harmonic

0	0
1	16
2	26
3	36
4	46
5	56

Add To Outputs

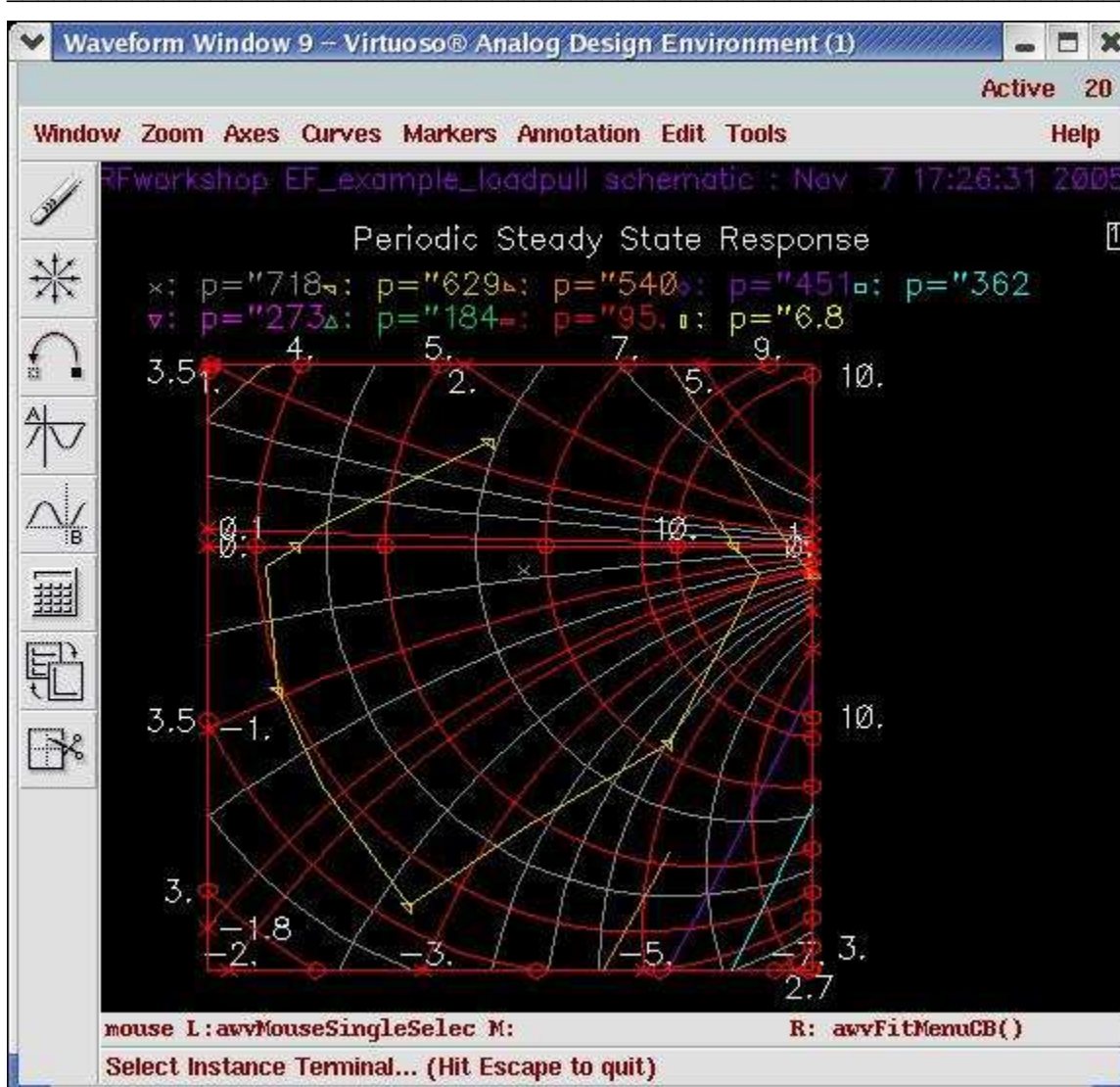
> Select Instance Terminal on schematic...

Action5-15: In the schematic window, select the **portAdapter** input terminal. If you want, click the **Close contours** button. The plot shows the contours of constant output power. The **X** at the center of the contours marks the optimal output power and its corresponding normalized impedance, Z_{opt} .



The small X appears at the maximum power point, which in this case lies near the center of the smallest constant power contour.

PA Design Using SpectreRF



If you place the cursor on the X, you can read the following information across the top of the Waveform window.

Real: 6.6376 Imag: -377.21m Freq: -360 p="718.768u"; Constant Power Contours

This indicates that a normalized load impedance of about $6.64 - j0.38$ dissipates the most power.

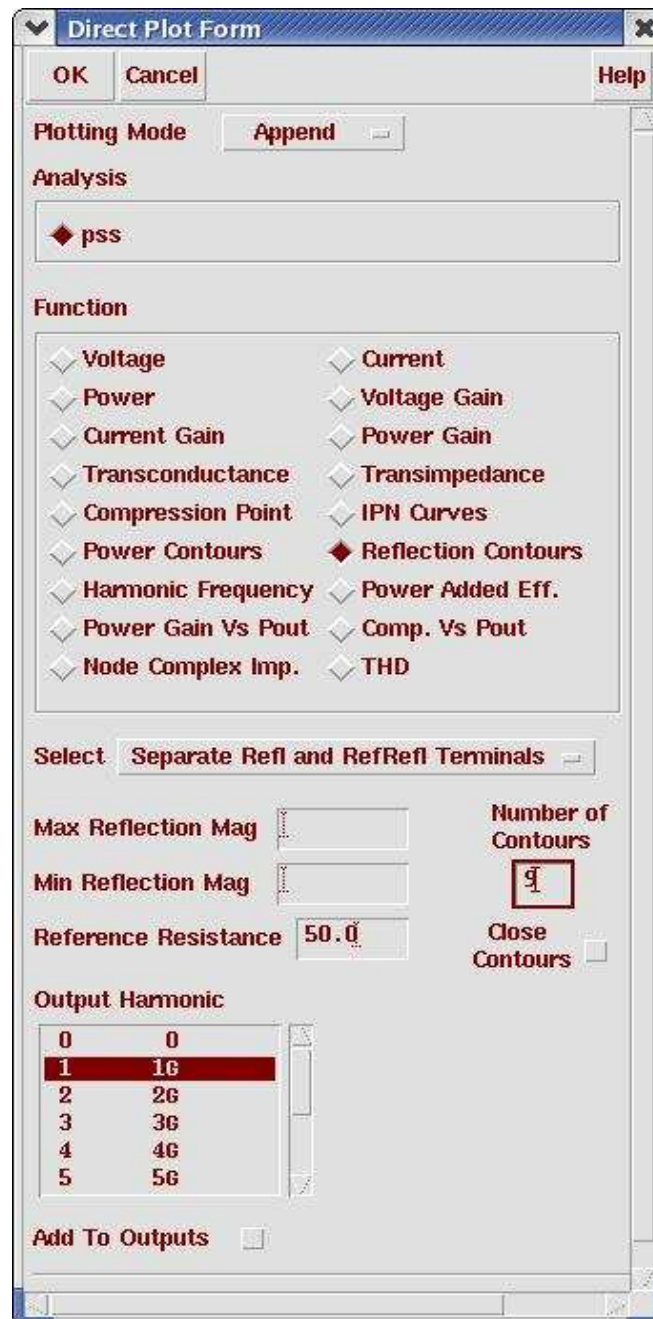
You might want to maximize load power subject to a constraint on the magnitude of the amplifier's input reflection coefficient. Such a constraint can prevent unstable interactions with the preceding stage.

You can overlay load-pull contours with contours of constant input reflection coefficient magnitude. The optimal load corresponds to the reflection coefficient that lies on the *largest power* load-pull contour and also lies on a constant input reflection coefficient

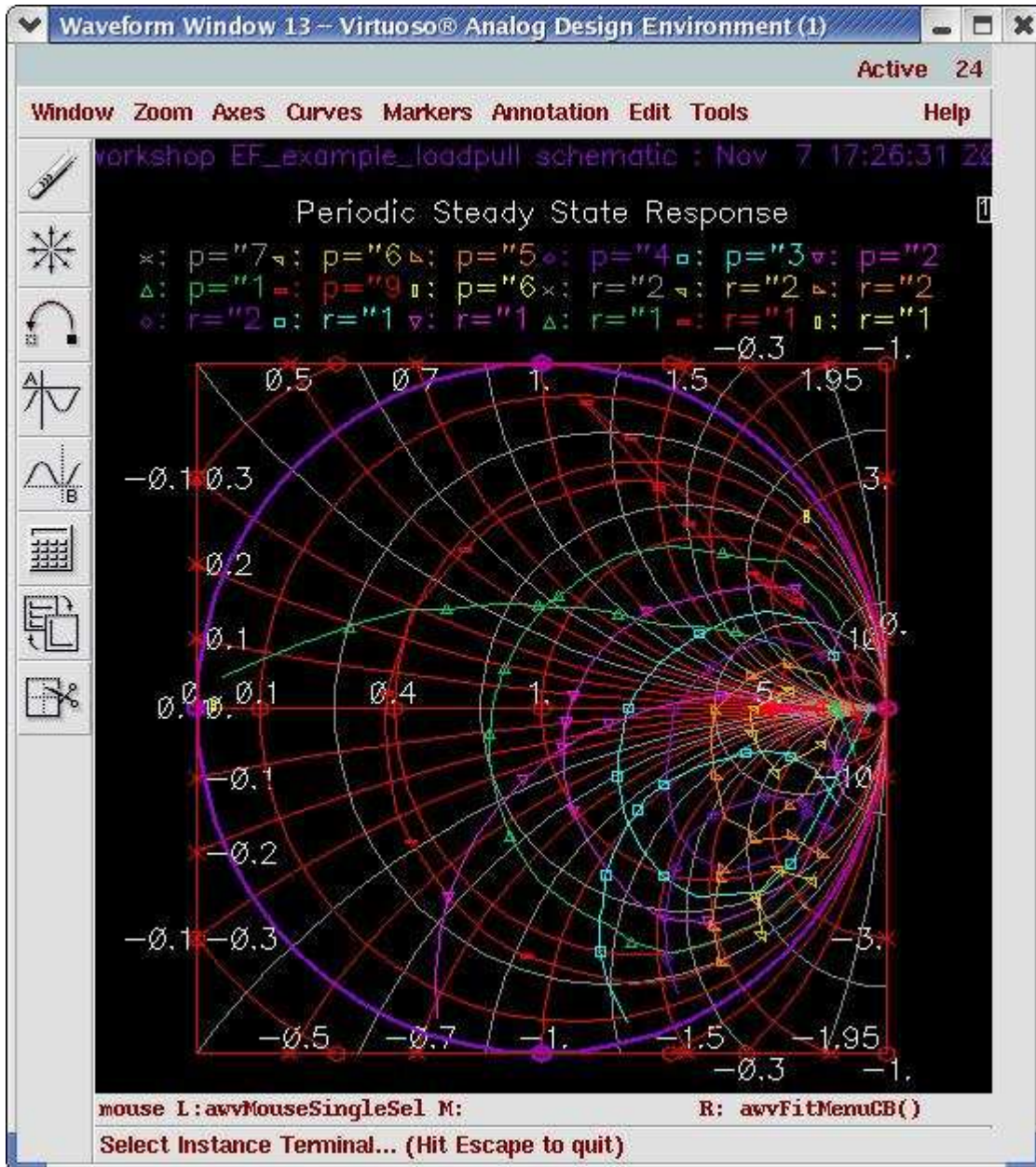
contour that is within the constraint. Here, *largest power* means the contour corresponding to the largest amount of power delivered to the load.

Action5-16: In the PSS Direct Plot form, choose the **Reflection Contours** function, then toggle **select** to **Separate Refl and RefRefl Terminals**. Select the PA's **input port (PORT1)** first, and then select the **portAdapter** input terminal. You are plotting the constant input reflection contours in the Smith chart of the output reflection coefficients.

The Direct Plot form should look like this:

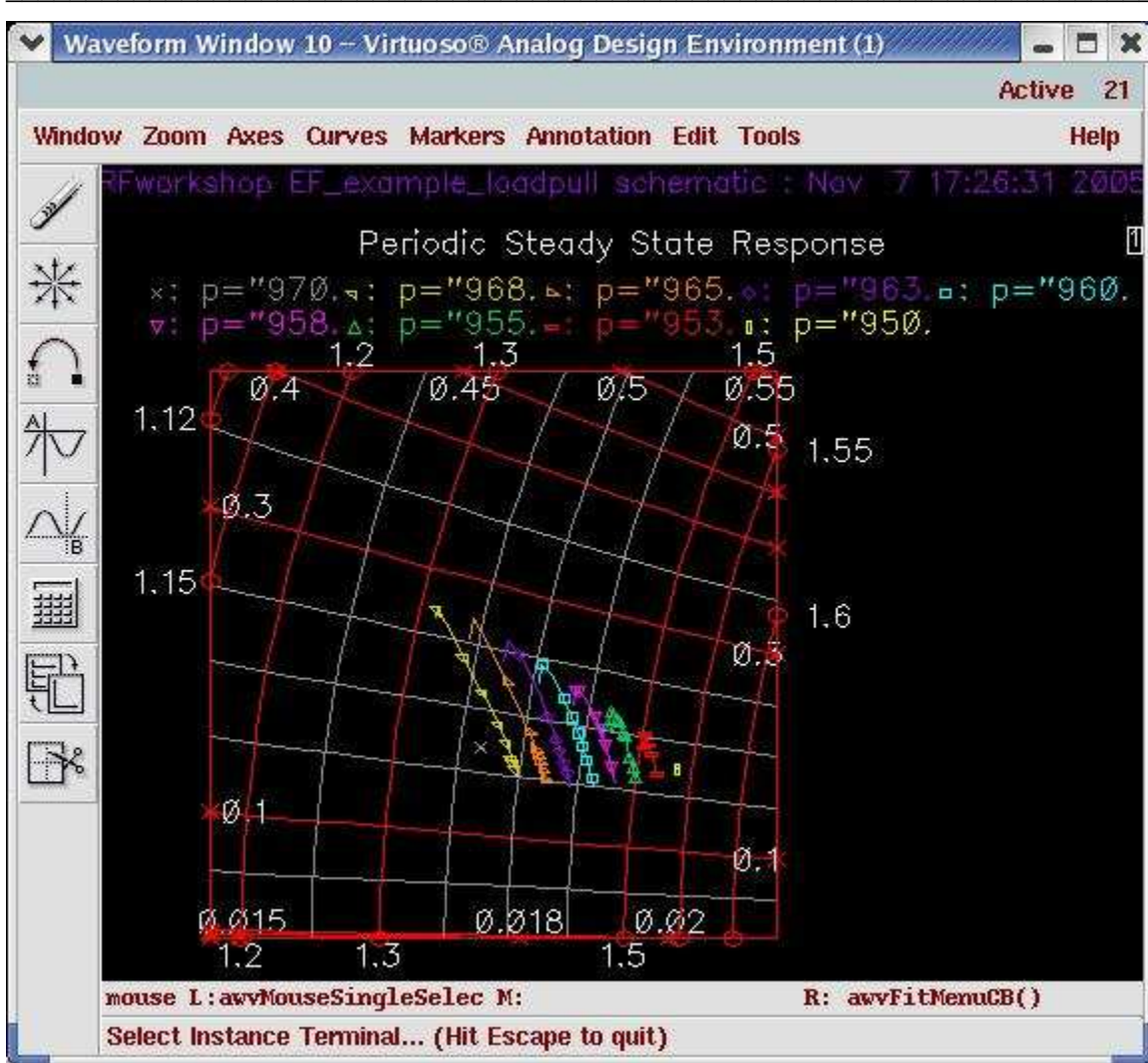


Here shows the constant input reflection coefficients contours overlying on top of the output power contour:



Action5-17: Changed the plot mode to **replace**, choose the **Power Contours** function, and select the terminal of the input port to plot the input power contour. If the contour shows that the input power does not vary significantly over the output reflection coefficient sweep, then the constant power contour is very close to the constant gain contour.

PA Design Using SpectreRF



Action5-18: Close the waveform window. Click **Cancel** on the Direct Plot form. Close the Virtuoso Analog Design Environment window.

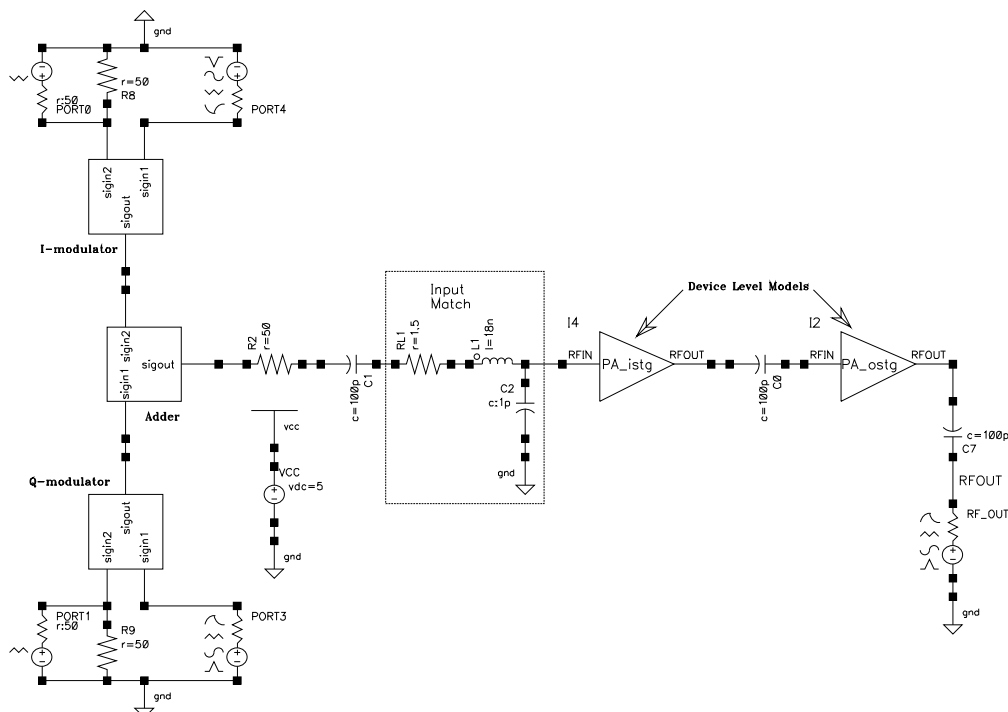
Lab6: Envelope Following Analysis (ENVLP and ACPR Wizard)

The envlp analysis is designed to generate an efficient and accurate prediction of the envelope transient response of circuits to different modulation schemes. The circuits are generally clocked at a frequency with a period that is orders of magnitude smaller than the baseband modulation signal. A classical transient approach is too expensive, and neither PSS nor QPSS work because the modulation signal is neither periodic nor quasi-periodic.

Envelope following analysis reduces simulation time without compromising accuracy by exploiting the behavior of circuits to a fixed high frequency clock. In particular, the envelope of the high-frequency clock can be followed by accurately computing the circuit behavior over occasional cycles. This accurately captures the fast transient behavior. The slow varying modulation cycle is accurately followed by a piecewise polynomial.

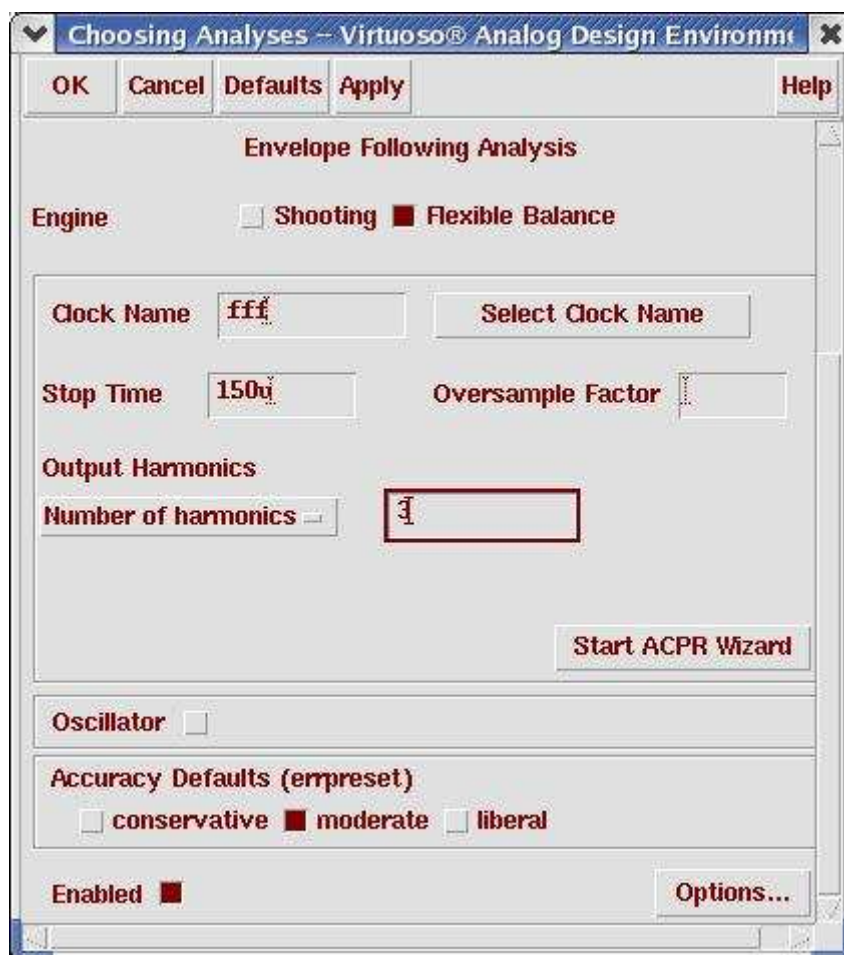
To do envlp analysis, you can use either shooting engine or Flexible Balance engine. This lab shows you how to use the Virtuoso® Spectre® RF Envelope with Flexible Balance engine to design and analyze transmitters.

Action6-1: Open the *schematic* view of the design *EF_example_envlp* in the library *RFworkshop*



The power amplifier is driven by modulation signals. CDMA I/Q baseband chip streams are fed into an ideal QPSK modulator.

- [Action6-2](#): View the object properties on PORT0 and PORT1. Note that the PWL file name for PORT0 is set to *cdma_2ms_idata*, and the PWL file name for PORT1 is set to *cdma_2ms_qdata*.
- [Action6-3](#): Check and save the schematic.
- [Action6-4](#): From the schematic window, start the Virtuoso Analog Design Environment with the **Tools—Analog Environment** command.
- [Action6-5](#): You can choose **Session—Load State**, load state “**Lab6_ENVLP_FB**” and skip to [Action6-12](#) or ...
- [Action6-6](#): In the Virtuoso Analog Design Environment window, click the **Choose Analyses** icon. The Choosing Analyses form appears.
- [Action6-7](#): Select the **envlp** analysis and choose the Flexible Balance engine. Set the Clock Name to **fff**, the Stop Time to **150u**, and the Number of harmonics to **3**. 3 harmonic is enough for this case. If the circuit is strongly nonlinear, you should choose more harmonics. If the circuit has square carrier, for some cases, 9 is acceptable, for some cases 20 or more should be used.
- [Action6-8](#): Set the Accuracy Defaults (errpreset) field to **moderate**.



Action6-9: Click the **Options** button at the bottom of the Choosing Analyses form.

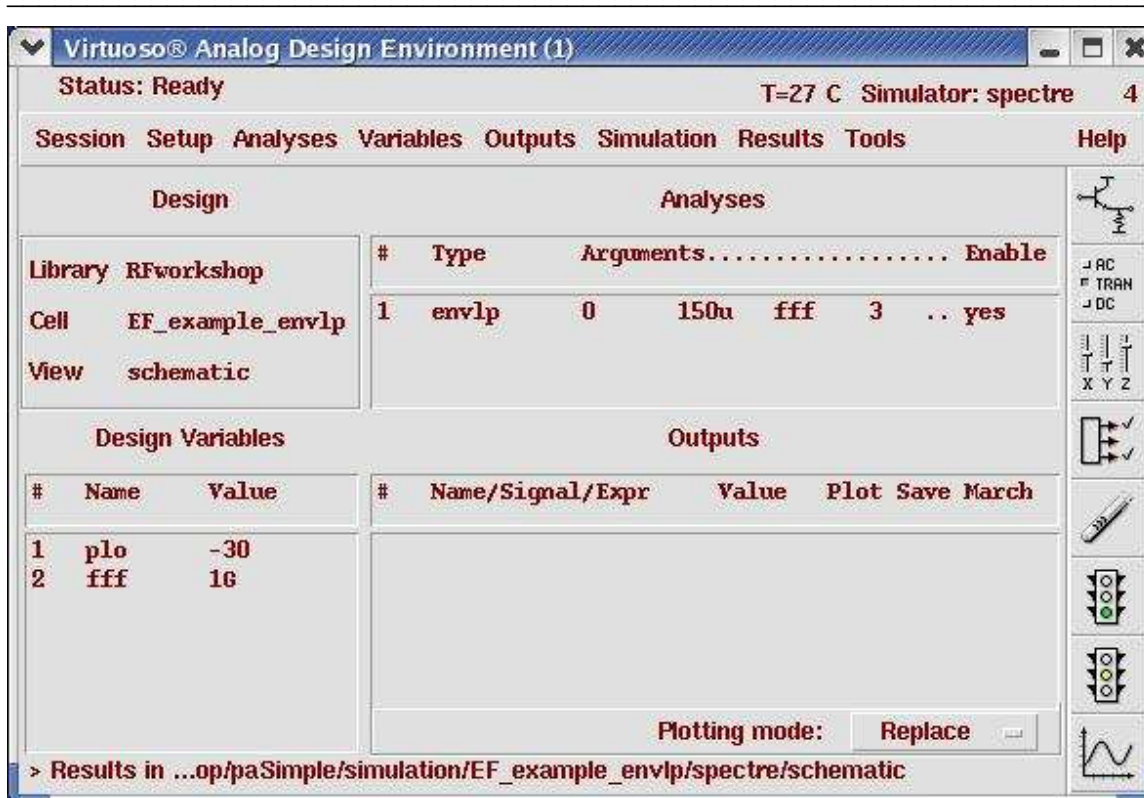
The Envelope Following Options form appears.

Action6-10: Under SIMULATION BANDWIDTH PARAMETERS, set *modulationbw* to **1M** (Hz) for this simulation.

Action6-11: Click **OK** in the Envelope Following Options form and then **OK** in the Choosing Analyses form.

The Virtuoso Analog Design Environment window should look like this:

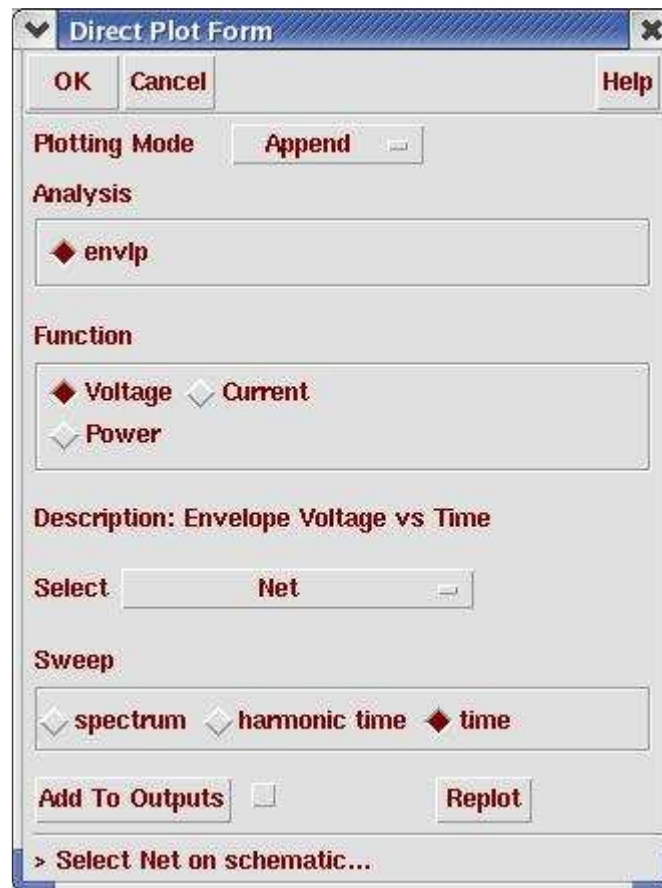
PA Design Using SpectreRF



Action6-12: In your Analog Design Environment, Choose **Simulation—Netlist and Run** or click the **Netlist and Run** icon to start the simulation.

Action6-13: In the Virtuoso Analog Design Environment window, choose **Results—Direct Plot—Main Form**.

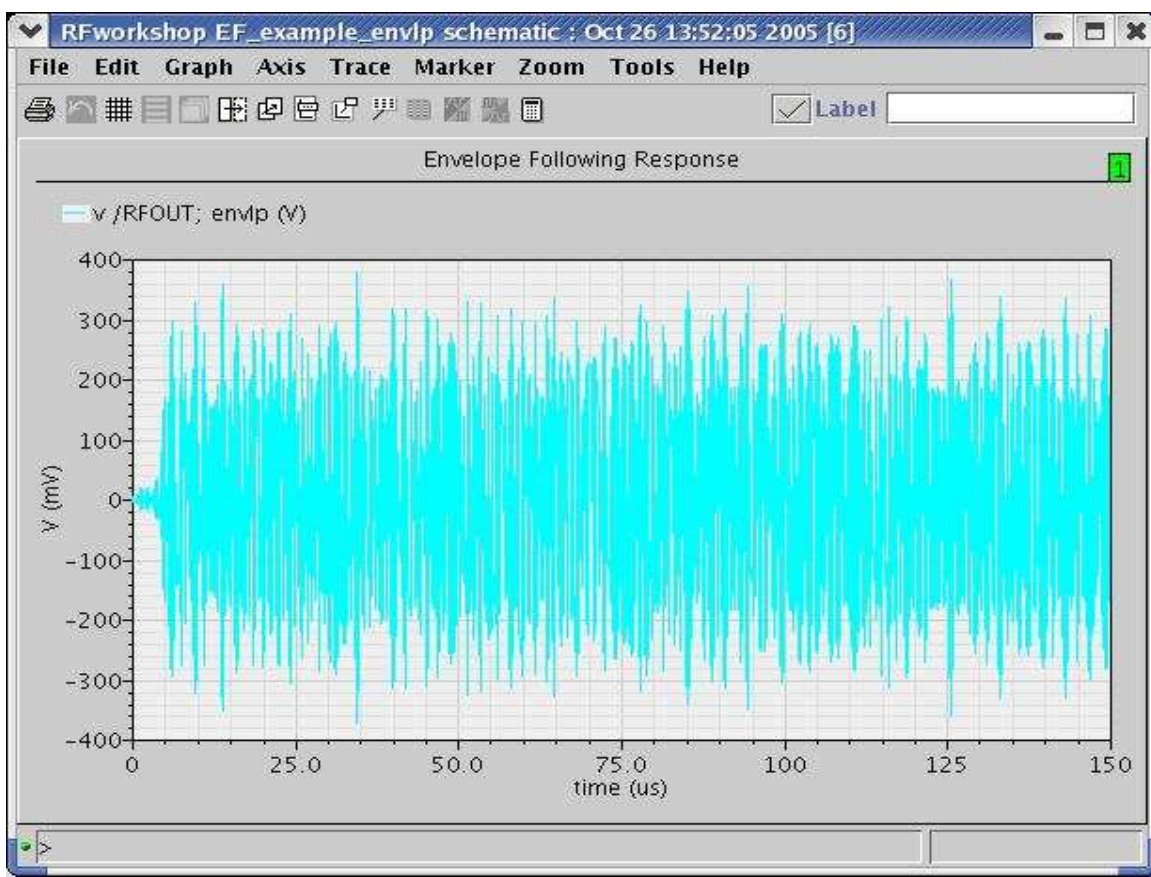
Action6-14: Select **Voltage** for Function. Select **time** for Sweep.



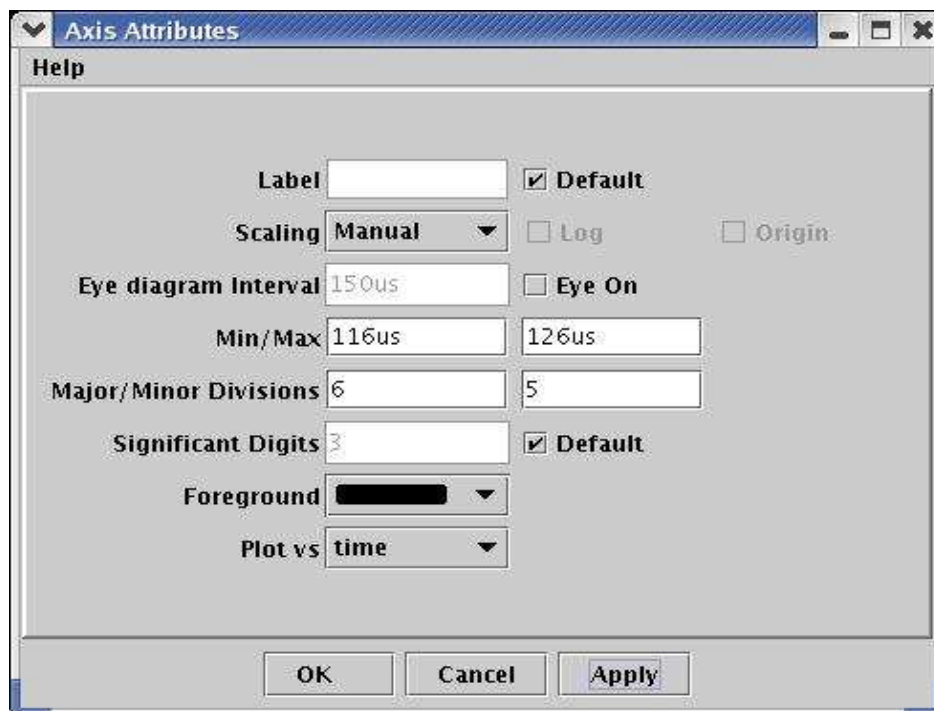
[Action6-15](#): In the schematic window, click on the *RFOUT* net.

The voltage waveform appears in the Waveform window.

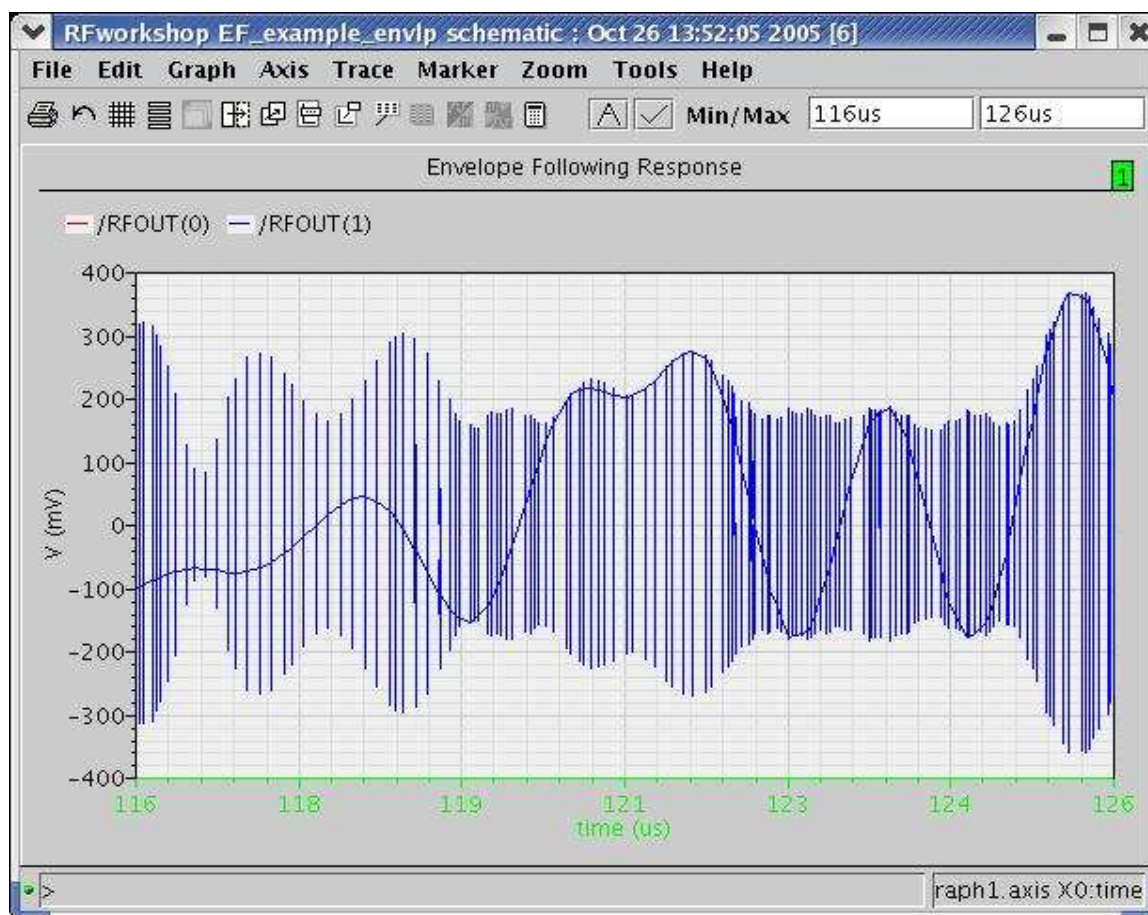
PA Design Using SpectreRF



Action6-16: In the Waveform window, double click on X-axis and set the Min and Max values as shown below.



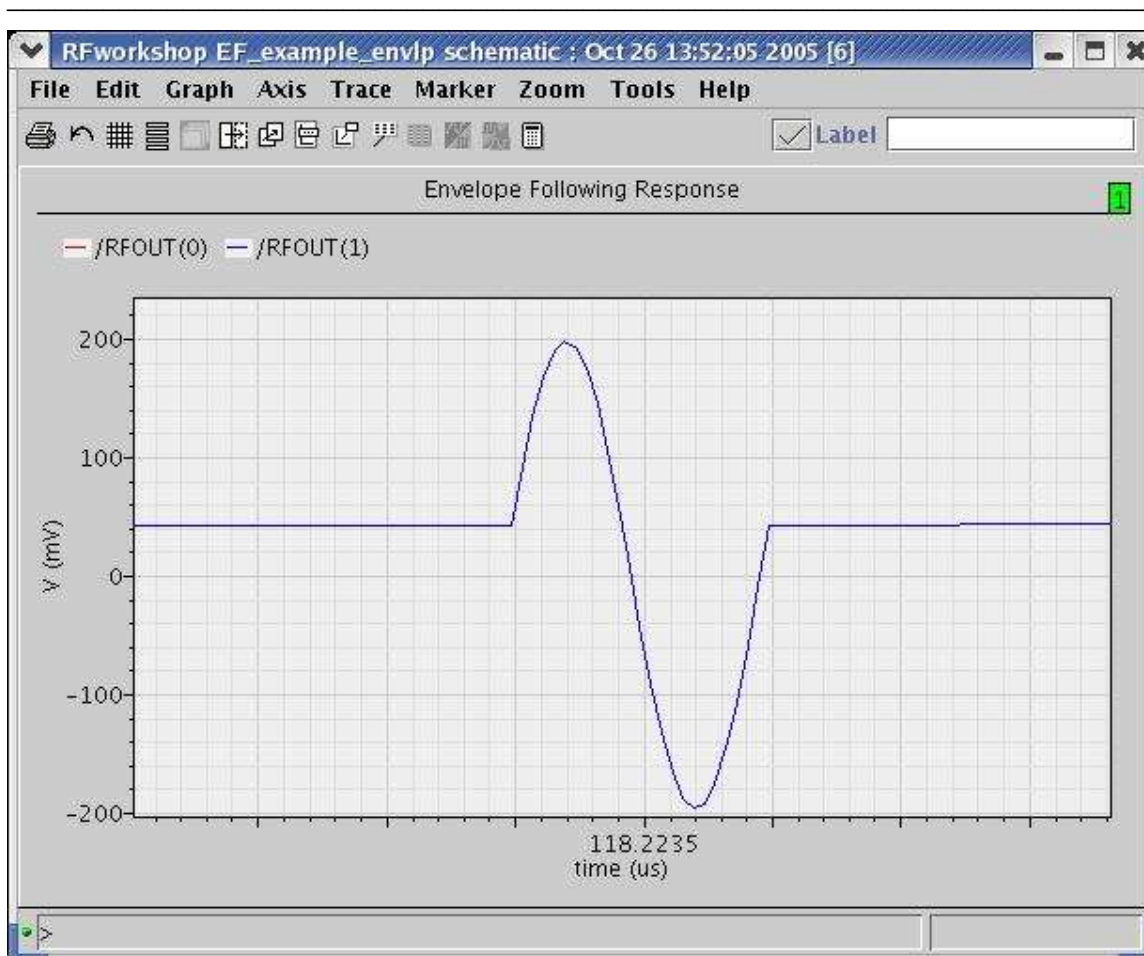
The Waveform window appears as follows.



The plot displays a number of vertical lines with a wavy line running through them. The vertical lines are the points at which detailed calculations are performed and the wavy line connects these points. The simulation runs much faster than a Virtuoso Spectre Transient Analysis simulation because Envelope Following skips carrier cycles when it can do so and still satisfy numerical tolerances.

Action6-17: To get a closer look, zoom in on any one of the vertical lines. You can now see the detailed simulation for one complete cycle.

PA Design Using SpectreRF



The modulation riding on the RF carrier is the baseband signal, the information to be transmitted. The baseband signal determines the amplitude and phase of the RF carrier. It is important to determine how the transmitter might alter the baseband signal. You can extract the baseband signal at any point in the design.

Action6-18: In the Direct Plot form, set these options:

- Select **Replace** for Plot Mode.
- Select **Voltage** for Function.
- Select **harmonic time** for Sweep.
- Select **Real** for Modifier.
- Select **1** for Harmonic Number.

Action6-19: In the schematic, click on the **adder** output.

A plot for the real portion appears in the Waveform window.

Action6-20: In the Direct Plot form, select **Append** for Plot Mode and **Imaginary** for Modifier.

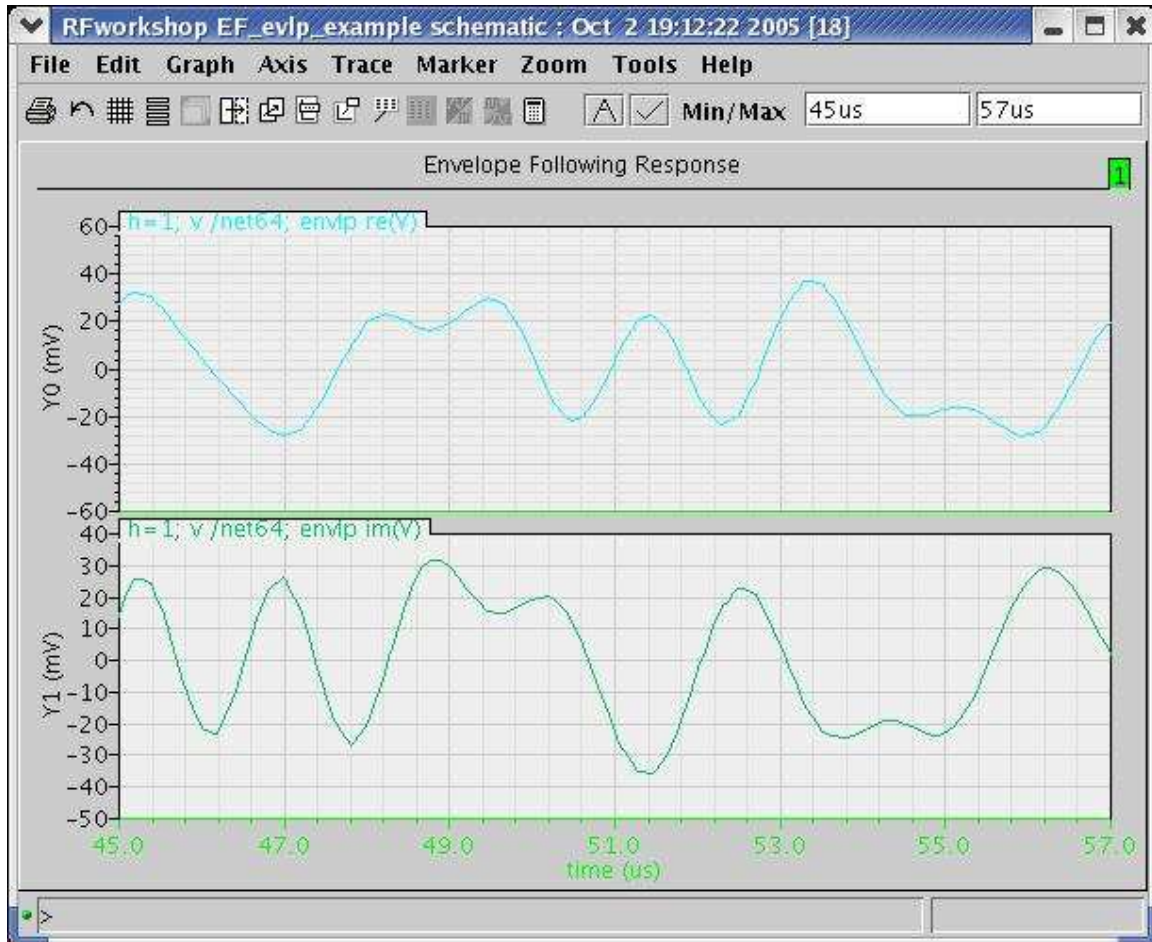
Action6-21: In the schematic, click on the adder output.

PA Design Using SpectreRF

A plot for the imaginary portion is added to the Waveform window.

Action6-22: In your waveform window, click on the Strip Chart Mode icon 

Action6-23: Set your X Axis to 45u to 57u, you can see both the real and imaginary parts clearly.



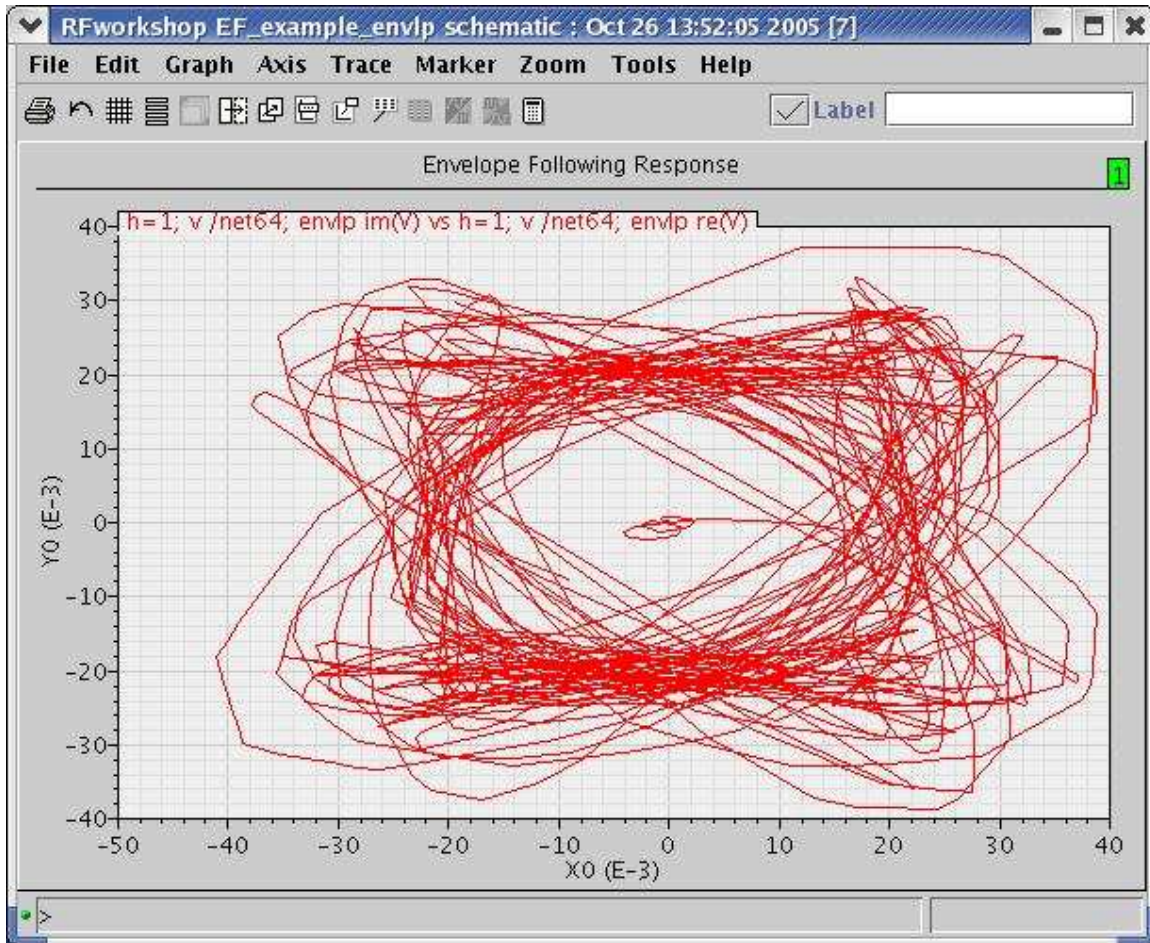
The baseband waveforms recovered from the modulated RF carrier, as displayed in the figures above, do not directly reveal much about how the transmitter affects them. The steps below tell you how to display the associated trajectory, which is the plot of one waveform against the other. The trajectory reveals much more about what kind of distortion the transmitter introduced. The steps below first display the input baseband trajectory and then the output baseband trajectory. A comparison of the two trajectories reveals whether the power amplifiers in this example are really distorting the signal.

Action6-24: In the Waveform window, double click on **X Axis** and set the Range to **Auto**.

Action6-25: In the *Plot vs. field* (at the bottom of the form), select **/net64 Voltage ; reV ; Harm = 1** and click **OK**.

PA Design Using SpectreRF

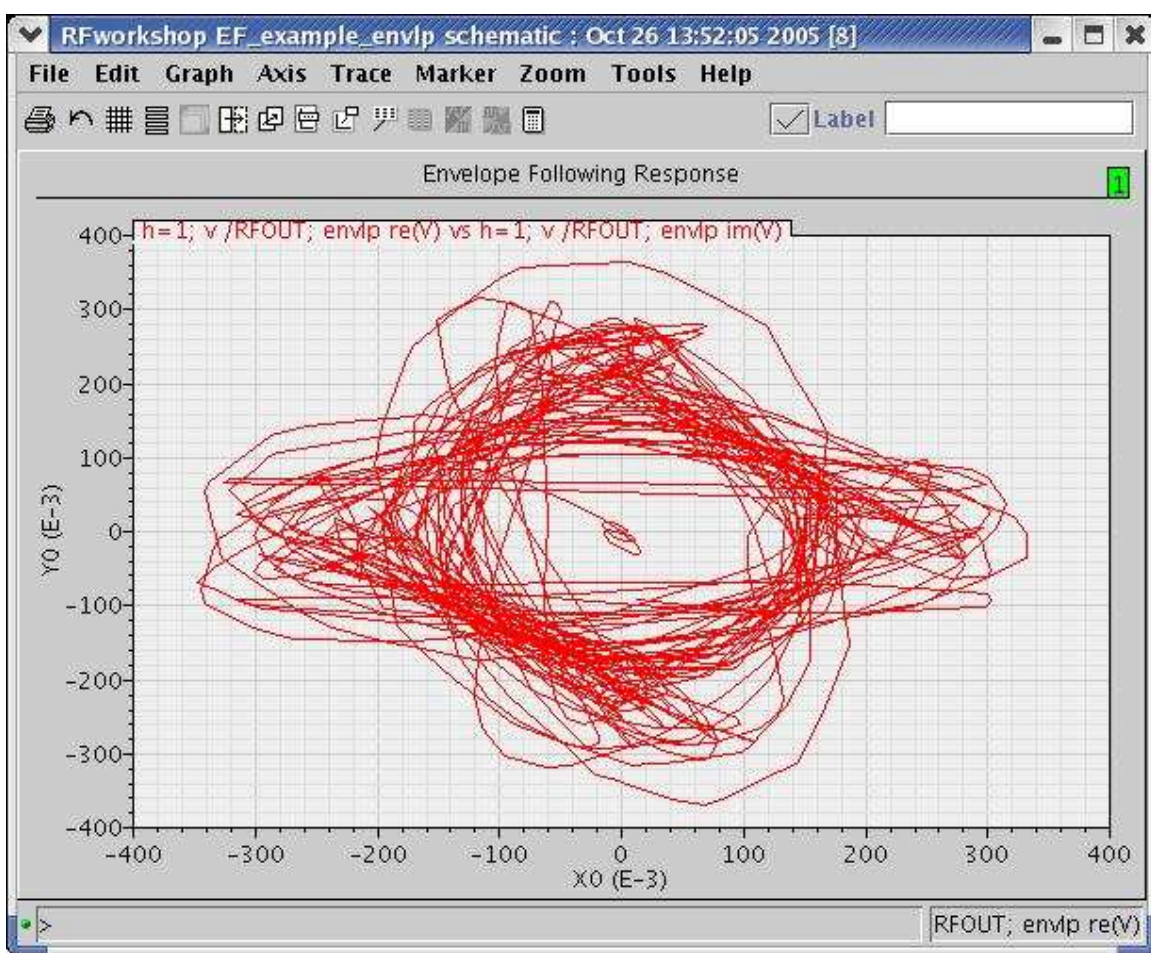
The plot below appears in the Waveform window. This is the input baseband trajectory, undistorted by the power amplifiers.



Action6-26: Close the Waveform window, then repeat the steps that you used to display the plot for `/net 64`, but substitute the `/RFOUT` net for `/net64`.

The plot you create in the Waveform window will look like this.

PA Design Using SpectreRF



The entire trajectory is scaled linearly and rotated. The output baseband signal is the input baseband signal, multiplied by a complex constant. The input and output *waveforms* look different because of the rotation, not because of some non-linear distortion. A common non-linear distortion, such as saturation, makes the outer edges of the trajectory lie on a circle.

The adjacent channel power ratio (ACPR) is a common index of how much power a transmitter emits outside its allotted frequency band. To measure ACPR, first obtain the power spectral density of the transmitted signal. This section describes how to plot the transmitted power spectral density.

To estimate ACPR, drive the transmitter with realistic baseband signals. In most cases, the baseband signals come from digital filters. The digital filters constrain the spectrum of the input baseband signal. Distortion in the transmitter causes the spectrum to grow where it should not. This growth is why you need an ACPR measurement.

The “uncategorized” part of the *rfLib* contains three sets of stored baseband waveforms, *cdma*, *dqpsk*, and *gsm*. These waveforms were created with the baseband signal generators in the testbench category of the *rfLib*. The *ppwlf* sources also read the System

Processing Worksystem (SPW) format. Therefore, these files can be generated using input baseband waveforms obtained through SPW.

- Action6-27:** In the Direct Plot form, set these options:
- a. Select **Replace** for Plot Mode.
 - b. Select **Voltage** for Function.
 - c. Select **spectrum** for Sweep.
 - d. Select **dB10** for Modifier.
 - e. Specify **1** for Harmonic Number.
 - f. Specify the Time interval from **0** to **0.0001**.
 - g. Type **5M** for Nyquist half-bandwidth.
 - h. Type **0.1M** for Frequency bin width.
 - i. Type **3M** for Max. plotting frequency.
 - j. Type **-3M** for Min. plotting frequency.

The completed form looks like this.

Direct Plot Form

OK Cancel Help

Plotting Mode Append

Analysis

◆ envlp

Function

◆ Voltage ◇ Current
◇ Power

Description: Harmonic Voltage Spectrum

Select Net

Sweep

◆ spectrum ◇ harmonic time ◇ time

Modifier

◇ Magnitude ◆ dB10 ◇ dBm

Harmonic Number

0
1
2
3

Power Spectral Density Parameters

Time Interval

From 0 To 0.0001 Get From Data

Nyquist half-bandwidth 5M

Frequency bin width 0.1M

Max. plotting frequency 3M

Min. plotting frequency -3M

Windowing Hanning

Detrending None

Add To Outputs Replot

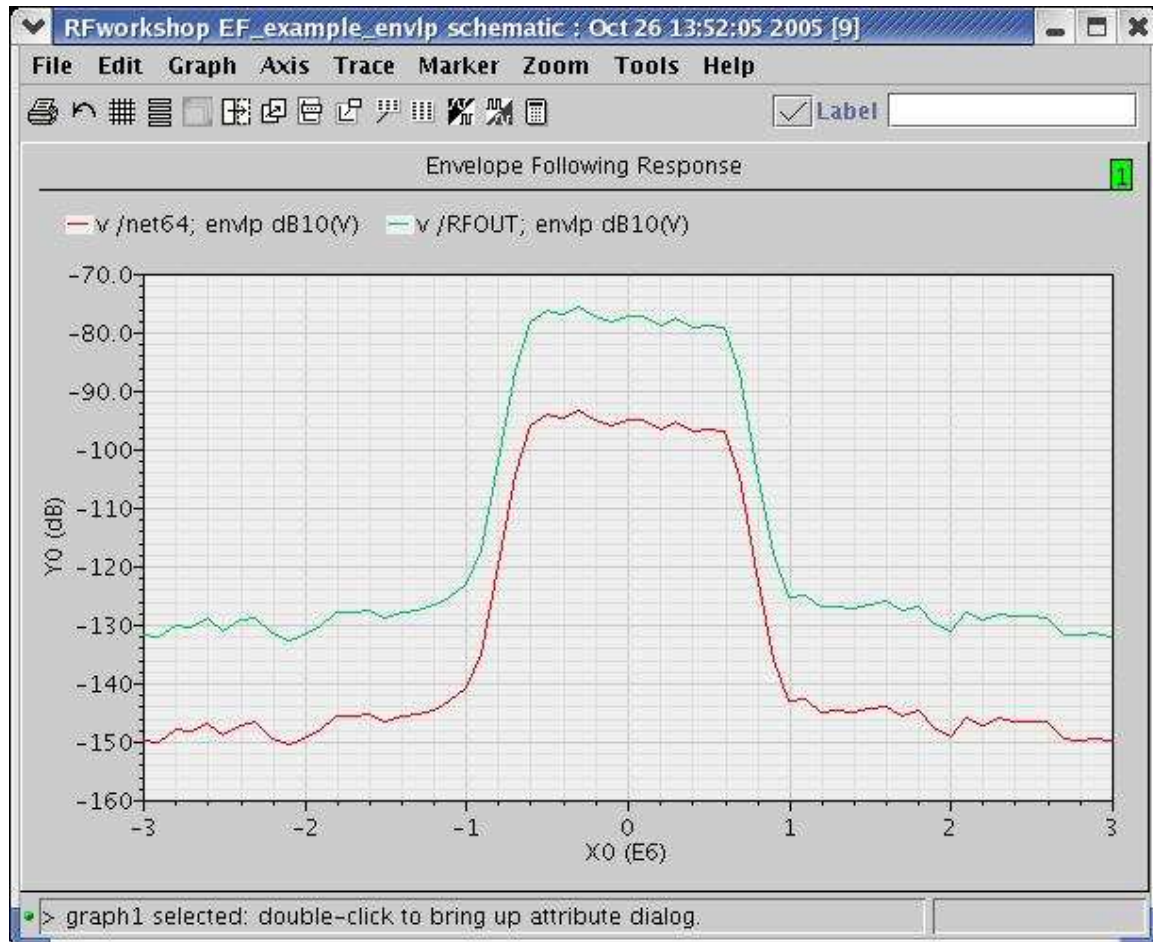
> Select Net on schematic...

PA Design Using SpectreRF

The stored waveforms for the input baseband signals were sampled at just under 5 MHz and the Nyquist half-bandwidth is also 5 MHz. This means the spectral algorithm must interpolate more than usual to generate enough time points for requested analysis.

The results in this case appear reasonable below 3 MHz but not beyond. As a general rule, keep the Nyquist half-bandwidth value below half the sample rate used to generate the input data. This example stretches the Nyquist criterion.

Action6-28: In the schematic, click on the RFOUT net and the output of the adder.



As you can see in the above figure, because the input level is very low, the PA is still working in linear region, the output power doesn't have too much leakage into the adjacent channel.

Action6-29: Calculate the ACPR for two x-axis values by subtracting their associated y-axis values. (ACPR measured with respect to x_1 and x_2 is $y_1 - y_2$).

Action6-30: Close the waveform window.

Using the ACPR Wizard

In this quick exercise, you will rerun the previous ACPR demonstration using the Spectre RF ACPR Wizard.

Action6-31: Open the ACPR wizard in one of two ways.

- In the Simulation window, choose **Tools - RF - Wizards - ACPR**
or
- In the **envlp** Choosing Analyses form, press **Start ACPR Wizard**.

In either case the ACPR Wizard displays.

Action6-32: Set the following:

Clock Name	fff
Net	/RFOUT
Channel Definitions	IS-95
Main Channel Width	5M
Stabilization Time	0
Resolution Bandwidth	7500 (calculate button)
Repetitions	2

The number of repetitions is set to 2, which gives a reasonable simulation time and accuracy. Increasing the number of repetition will provide a better accuracy at a cost of a longer simulation time.

Your ACPR wizard form should look like this:

The ACPR Wizard dialog box is shown with the following settings:

- Buttons:** OK, Cancel, Apply, Help
- Clock Name:** fff
- How to Measure:** Net
- Net:** /RFOUT
- Select:** [button]
- Channel Definitions:** IS-95
- Main Channel Width (Hz):** 5M
- Adacent frequencies are specified relative to the center of main channel**
- Table:**

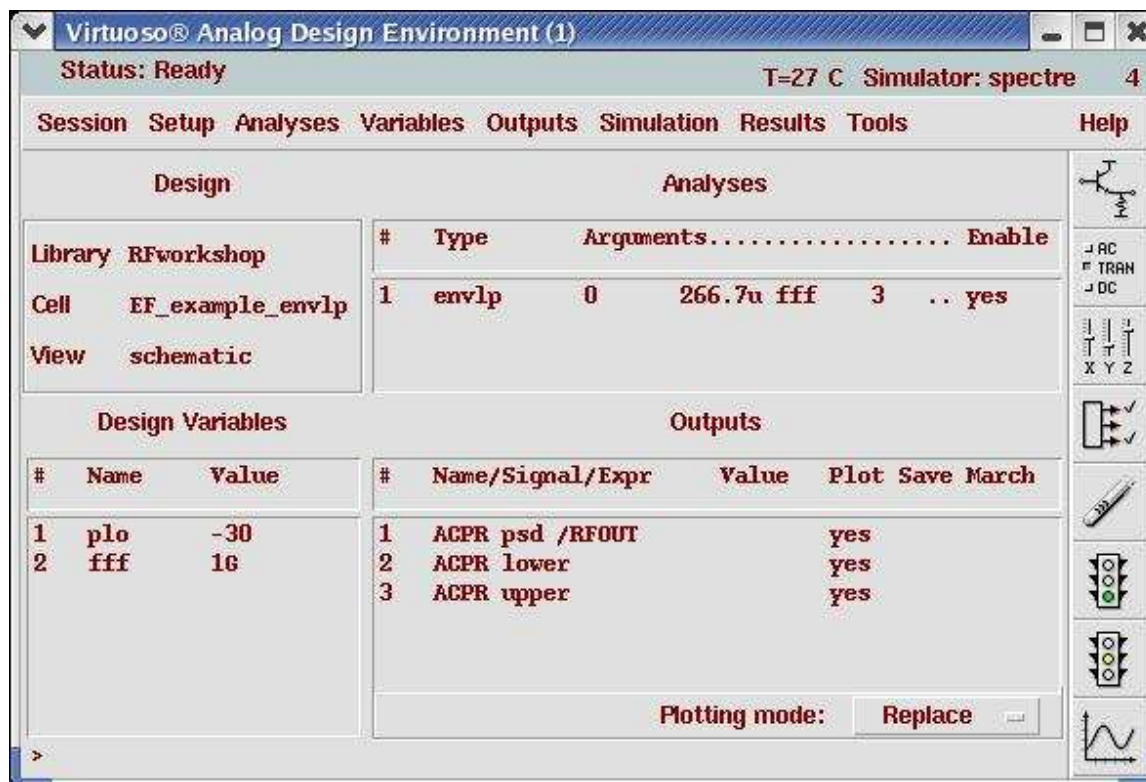
name	from (Hz)	to (Hz)	
lower	-915K	-885K	
upper	885K	915K	
- Buttons:** Add, Change, Delete
- Simulation Control:**
 - Stabilization Time (Sec):** 0
 - Resolution Bandwidth (Hz):** 7500 [Calculate button]
 - Repetitions:** 2
- Windowing Function:** Hanning

Action6-33: On the ACPR Wizard form click **Apply**.

This action loads the output section of the ADE window with your selected values.

PA Design Using SpectreRF

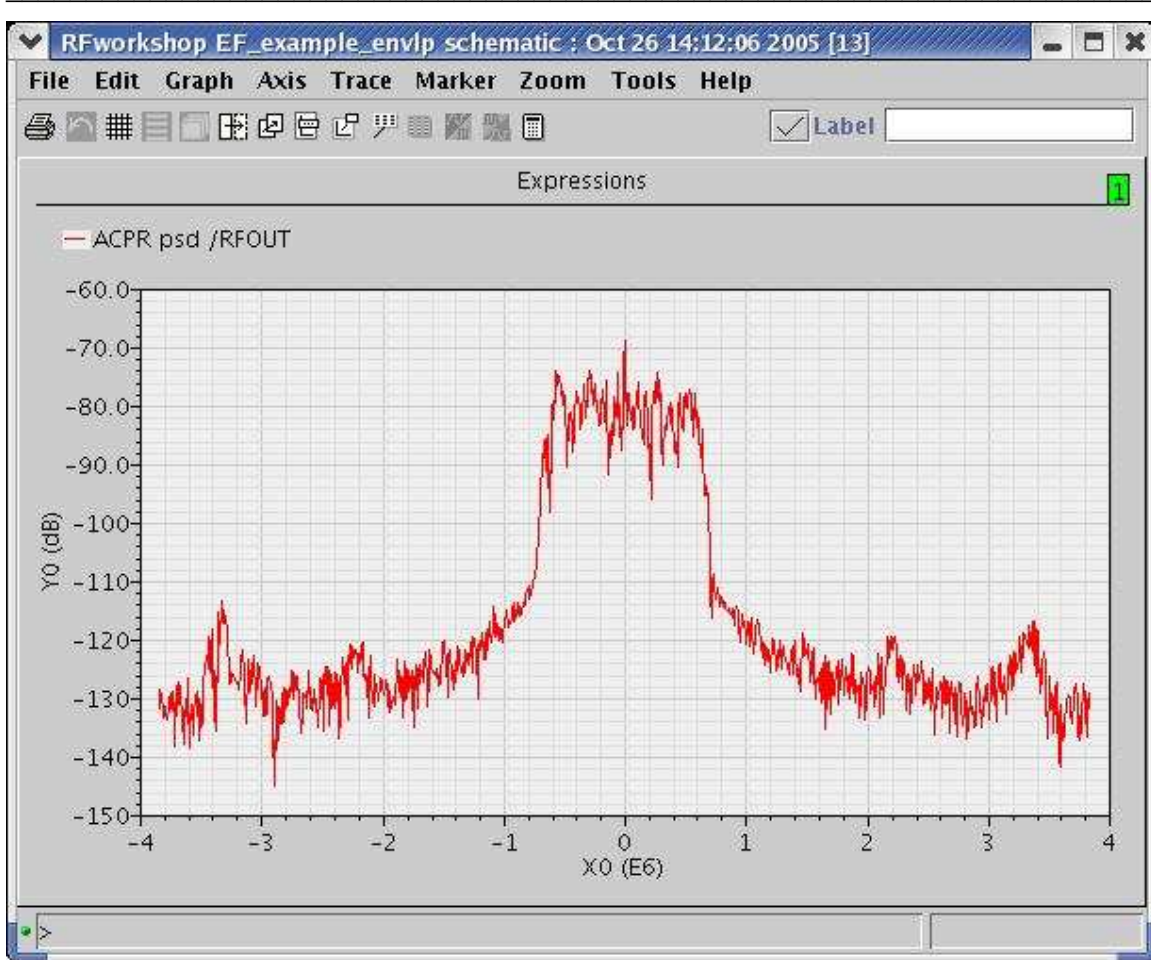
The ADE window now looks like this:



Action6-34: In the ADE window, press the plot button on the right-hand toolbar.

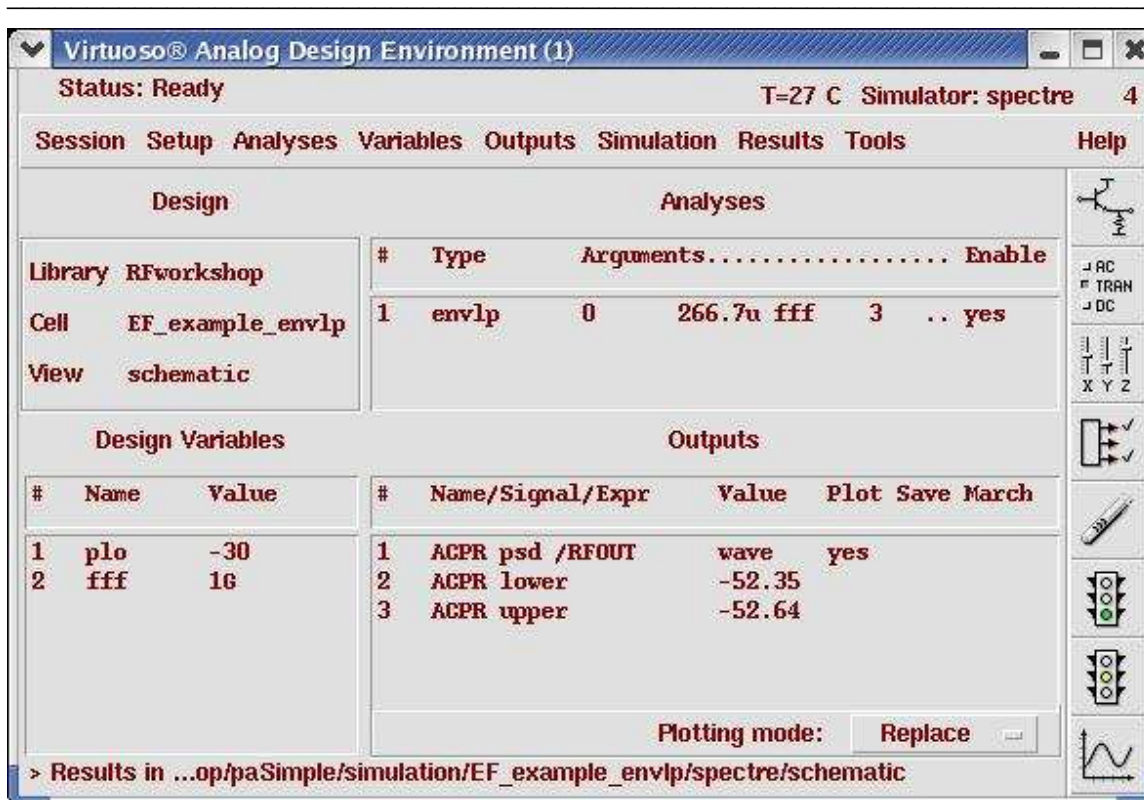
The Waveform Window opens:

PA Design Using SpectreRF



Now your Virtuoso Analog Design Environment window look likes this:

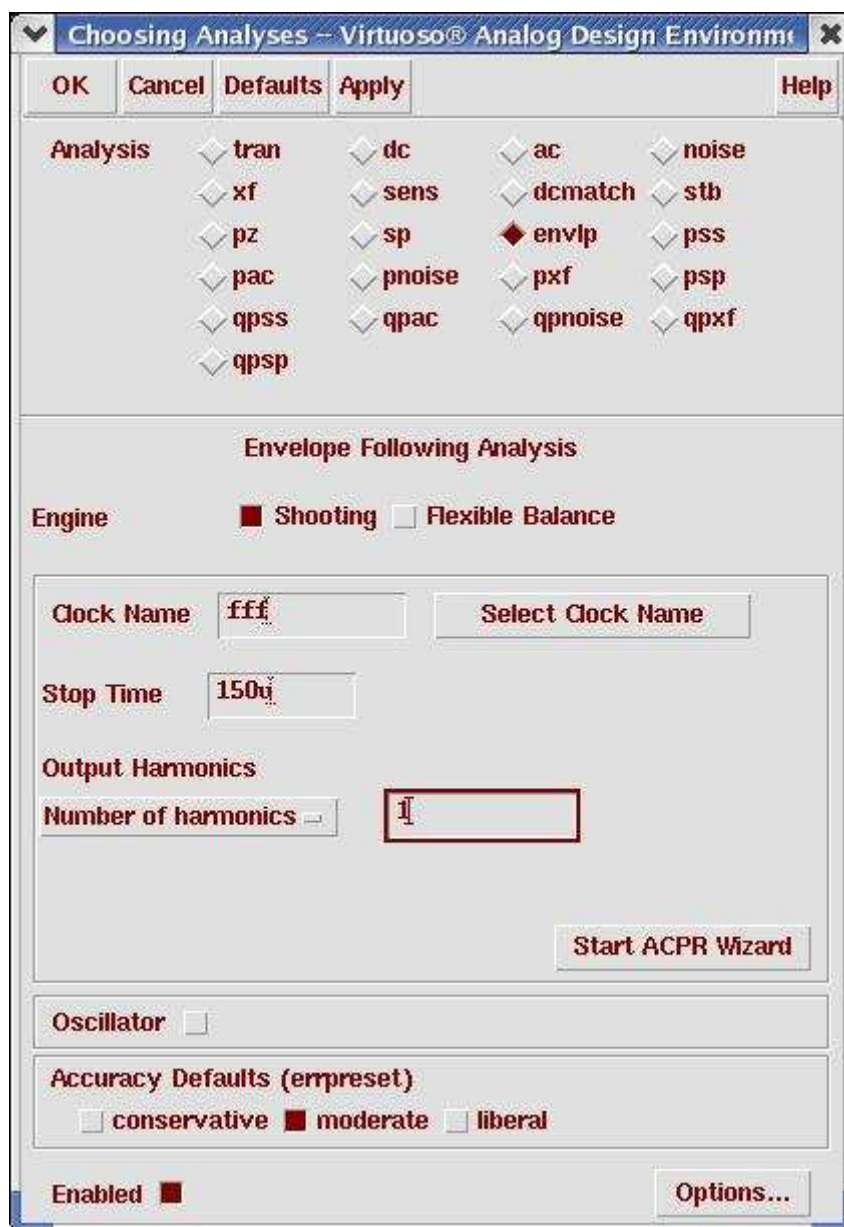
PA Design Using SpectreRF



[Action6-35](#): Rerun these steps a few times substituting values in the ACPR Wizard.

You can change the Flexible Balance engine to shooting engine, change the number of harmonics to 1, and re-run the simulation. Or you can load the state “Lab6_ENVLP_shooting” and repeat [Action6-12](#) to [Action6-30](#).

The envlp analysis form with shooting engine will look like:



You can also change the plo level and re-run simulation with both Flexible Balance engine and shooting engine. You will conclude that for linear or weakly nonlinear circuit, Flexible balance engine is faster.

Action6-36: Close the waveform window. Click **Cancel** on the Direct Plot form. Close the Virtuoso Analog Design Environment window.

Conclusion

This workshop describes how to use spectreRF for RF power Amplifiers designs. It first presents the typical PA design parameters and describes how to build testbenches and perform measurements within Analog Design Environment. It then covers in detail how to set up spectreRF analyses and perform measurements related to PA design. Lastly, this workshop displays and interprets the simulation results.

Reference

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- [2] T. Lee, “The Design of CMOS Radio Frequency Integrated Circuits”, Cambridge University Press, 1998.
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- [4] M. Hella, “RF CMOS Power Amplifiers: Theory, Design and Implementation”, Kluwer Academic Publishers, 2002.